MILITARY SPECIFICATION

STARTER, ENGINE, CARTRIDGE/PNEUMATIC, SHAFT DRIVE, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for cartridge/pneumatic starters for aeronautical gas turbine engines.

1.2 Classification. The type and model designation will be assigned by the procuring activity in accordance with MIL-STD-875 and shall be specified in the model specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-T-5021	Tests; Aircraft and Missile Welding Operators' Qualification
MIL-C-6021	Castings, Classification and Inspection of
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-C-8188	Corrosion-Preventive Oil, Gas Turbine Engine, Aircraft Synthetic Base
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-S-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electrical Systems, Accelerator Required

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MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-23586	Sealing Compound, Electrical, Silicone Rubber, Accelerator Required
MIL-W-25038	Wire, Electrical, High Temperature and Fire Resistant, Aircraft
MIL-E-25499	Electrical Systems, Aircraft, Design and Installation of, General Specification for
MIL-C-27505	Cartridge, Engine Starter MXU-4A/A
MIL-C-27658	Cartridge, Engine Starter MXU-129/A
MIL-V-38398	Valve, Starter Control, Aircraft Engine, General Specification for
MIL-I-81550	Insulating Compound, Electrical, Embedding, Reversion Resistant Silicone

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-453	Inspection, Radiographic
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-490	Specification Practices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

438	Age Controls of Age-Sensitive Elastomeric Items

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Model specification. When specified (see 6.2), a starter model specification conforming to the appendix of this specification shall be prepared by the contractor and submitted to the procuring activity for approval (see 6.3.24). The model specification shall provide all the necessary information to completely describe the cartridge/pneumatic starter for one weapon system application. Approval of the model specification is required prior to initiation of preproduction and quality conformance testing. Changes and revisions to the model specification shall be in accordance with MIL-STD-490.

3.2.1 Detail specification. Requirements for individual items shall be as specified herein and shall be in accordance with the applicable detail specification (see 6.3.25) for the system.

3.3 Mockup. A full-scale mockup shall be prepared for examination and aircraft installation suitability as soon as the contractor has established the necessary design features of the starter. Prior to initiation of the preproduction testing, the mockup shall be installed on the aircraft for which the design was finalized to demonstrate satisfactory fit and installation features. The mockup shall require approval of installation suitability by the procuring activity.

3.3.1 Changes. Changes to the starter features requiring changes in the vehicle or ground support equipment made after approval of the mockup shall be submitted to the procuring activity for approval in accordance with MIL-STD-480. Any changes required by the procuring activity shall be subject to negotiation as provided in the contract. Unless otherwise authorized, the mockup shall be kept current with approved changes at least through the first production contract.

3.4 Components. Normally, the starter shall consist of the following assemblies and systems:

a. Breech and gas transfer assembly 3.9.1

b. Energy conversion assembly 3.9.2

c. Speed reduction assembly 3.9.3

d. Clutch and output assembly 3.9.4

e. Control system 3.9.5

f. Duct system 3.9.6

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g. Power system 3.9.7

h. Lubrication system 3.9.8

3.5 Selection of specifications and standards. Specifications and standards for necessary items and services not specified herein shall be selected in accordance with MIL-STD-143.

3.6 Materials

3.6.1 Metals. Metal parts shall be of the corrosion-resistant type or suitably treated to resist corrosion due to cartridge combustion gases or residue, or atmospheric conditions likely to be met in storage or service use. Any corrosion-resistant protective coating applied shall provide maximum protection against cracking, chipping, or scaling of the coating with age, use, or extremes of environmental conditions.

3.6.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.6.2 Nonexpendable metal parts. Nonexpendable metal parts may be reworked and reused as agreed between the procuring activity and the contractor, if such parts comply with specifications and drawings in all respects.

3.6.3 Synthetic rubber parts. Age of synthetic rubber parts shall be controlled in accordance with ANA Bulletin 438.

3.6.4 Castings. Castings shall be in accordance with MIL-C-6021.

3.6.5 Potting compounds. Requirement 47 of MIL-STD-454 shall be adhered to for encapsulation and embedment. Only those materials which meet the requirements of MIL-S-8516, MIL-S-23586, or MIL-I-81550 shall be used.

3.6.6 Organic materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, they shall be treated with a suitable fungicidal agent.

3.7 Design and construction. The starter shall be designed for operational use with a cartridge and a pneumatic energy source, and its design shall be amenable to production processing methods. The starter shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service usage. The cartridge and pneumatic energy sources shall be defined in the detail specification and model specification.

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3.7.1 Overspeed protection. The starter shall be designed to incorporate both of the following features:

a. The starter shall be capable of containment of all starter fragments within its envelope and shall remain on its mount in the event an overspeed failure should occur under any operating condition. Turbine wheel hub burst containment is required to a speed not less than the maximum operating speed of the starter under normal operating conditions.

b. The starter shall incorporate a speed-limiting feature which will safely limit the starter speed, either loaded or unloaded, and shall not allow the starter to assist the engine beyond the engine's maximum starter cutout speed when supplied with the energy sources specified in the model specification. The speed-limiting feature shall not require resetting and shall function, as necessary, throughout the life of the starter without maintenance or adjustment. In addition, the speed-limiting feature shall be totally contained within the starter and shall not require special external power of any kind. Electrical power will be supplied from the aircraft electrical system during operation in the pneumatic mode in applications where a pressure-regulating and shutoff valve is incorporated in the aircraft system as well as momentary electrical power for cartridge actuation.

3.7.1.1 Auxiliary air. If the starter requires ingestion of auxiliary air for speed limiting or other applications, the inlet shall be protected against ingestion of foreign objects. The rate of flow required for the starter to function satisfactorily under normal operating conditions shall be specified in the model specification. The air may be taken from within the nacelle; however, the contractor must demonstrate that no fire hazard is created. If the air must be taken from outside the nacelle, the starter shall incorporate provisions for attachment of an airframe-furnished duct. The duct diameter and length required for satisfactory operation under extreme and normal operating conditions shall be specified in the model specification.

3.7.2 External surface temperature. The external surface temperature of the starter shall not exceed 700°F during or after any operating condition. This requirement shall be applicable for all modes of starter operation throughout the ambient temperature range specified in the model specification. Provisions necessary to meet this requirement shall be integral with the starter.

3.7.3 Arming. The starter shall be suitably protected against inadvertent ignition of the cartridge. It shall be possible to furnish electrical power to the cartridge only after securing the breech. It shall be possible to disarm the cartridge at any time prior to actuation.

3.7.4 Gas leakage. There shall be no external leakage of gases from any part of the starter during or following any operating condition.

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3.7.5 Interlock prevention. In the event a gearbox switch and a turbine lockout switch are used, their points of actuation (rpm at actuation) shall be sufficiently far apart to prevent interlocking of the gearbox and turbine. These points of actuation with tolerances and the amount of drift in the settings shall be specified in the model specification.

3.7.6 Emergency disengagement. A disengaging mechanism shall be incorporated in the starter to effect disengagement of the starter from the engine in the event normal disengagement does not occur. The disengaging mechanism shall be so designed that removal of the starter from the engine pad shall be required to effect re-engagement. After an emergency disengagement has occurred, an attempted start shall not result in damage to the starter or engine. The disengaging mechanism shall be described in detail and its operating limits specified in the model specification.

3.7.7 Flight and ground loading conditions. The starter shall withstand, without permanent deformation or failure, the largest forces resulting from all critical combinations of loads and rotational accelerations specified in 3.7.7.1. For design purposes, minimum yield strength shall provide for at least 1.5 times the largest forces resulting from the loading conditions and a minimum ultimate strength of at least 1.67 times the largest forces resulting from the loading conditions. The design limit torque load shall be the highest value of torque specified in the model specification and shall also include the torque loading conditions resulting from turbine wheel seizure during containment. For design purposes, the weight of the starter shall include the cartridge, ducting, and all components and parts that make up the complete starter assembly.

3.7.7.1 External loads. The load factors and rotational accelerations specified below are to be considered acting separately and in combination with the design limit torque load and overhung moment:

a. Applicable loads and accelerations resulting from maximum flight maneuvers and ground loading conditions of 10g, acting in any direction through the center of gravity of the starter, when the starter contains a fired cartridge combined with the design limit, pressure, torque, and overhung moment loads.

b. Duct attachment load. The starter duct connecting flange(s) shall be capable of withstanding, without permanent deformation or failure, a force of 50 pounds direct thrust acting either inward or outward or a moment of 150 pound-inches (lb-in.) acting in any direction around the axial centerline of the flange.

3.7.8 Overhung moment. Maximum overhung moment of the starter, filled with oil, including an unfired cartridge, shall be specified in the model specification and shall not exceed the value specified in the detail specification when measured from the face of the engine mounting flange. The starter center-of-gravity locations, weights, and overhung moments shall be specified on the

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starter vendor's outline drawing for the following conditions:

a. Starter filled with oil

b. Starter filled with oil and including a spent cartridge

c. Starter filled with oil and including an unfired cartridge.

3.7.9 Polar moment of inertia. The starter effective mass polar moment of inertia at the starter drive shall be specified in the model specification.

3.7.10 Reliability. The starter shall be designed and developed to achieve the highest operational reliability commensurate with the design requirements. The starter shall have a reliability of 0.998 based on a one-cycle mission with a confidence level of 90 percent. Successful completion of all tests in section 4 of this specification will have demonstrated a suitable reliability. Factors to be considered during the design and manufacture of the starter to assure in-service reliability are:

a. Simplicity of design

b. Selection and application of reliable components

c. Consideration of operational and environmental parameters

d. Mechanical structures - adequate design stress margin on all parts

e. Design review by technical specialists prior to drawing release

f. Use of proven manufacturing techniques

g. Rigid quality control procedures imposed throughout procurement, manufacturing, assembly, and testing to assure that maximum design reliability is built into the equipment.

3.7.11 Maintainability. The starter shall have a minimum number of parts consistent with reliability. Its design shall permit easy assembly, disassembly, location of trouble sources, routine service checking, adjustment, and maintenance by service maintenance personnel with a minimum of training using tools and equipment normally available commercially. Maintenance between overhaul periods shall be held to an absolute minimum with a design requirement of routine oil changes at 200 flight-hour intervals. The man minutes required to check and change oil and perform all other field maintenance shall be specified in the model specification. Maintainability considerations shall be in accordance with MIL-STD-470.

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3.7.12 Overhaul life. The unit shall perform in accordance with the requirements of this specification throughout 400 cartridge starts or 1,000 total starts, whichever occurs first, and overrunning (normal aircraft flight operating time) as required for the applicable installation and specified in the detail specification.

3.7.12.1 Recording device. The starter shall include a device which records cartridge firings. The device shall be mounted on the starter, be readable, and shall not be resettable in the field.

3.7.13 Threads

3.7.13.1 Threads. All conventional straight screw threads shall conform to MIL-S-7742.

3.7.13.2 Threaded inserts. All threaded connections in nonferrous materials which are heavily loaded or removed frequently shall have steel inserts which are suitably protected from electrolytic corrosion. Fill and drain boss inserts shall be designed to permit the use of standard gaskets or seals and standard straight thread fittings.

3.7.14 Starter adapter flange. Unless the engine mounting pad provides a quick attach-detach feature, an adapter shall be furnished as a part of the starter. This adapter shall be bolted to the engine drive pad and shall provide for torsional restraint engagement and quick attach-detach of the starter.

3.7.15 Inspection seals. All starter performance adjustment points shall be preset at the factory, and inspection seals shall be provided so that field adjustments cannot be made without breaking a seal.

3.7.16 Safetying. All fittings and attachments shall have approved self-locking features or shall be lock wired.

3.7.17 Drainage. The starter shall have provisions for draining or shall be sealed to prevent water from accumulating within the unit. Drainage shall be such that any condensed or accumulated water will not cause malfunction or cause delay in starting if frozen.

3.7.18 Fitting identification. The starter shall be permanently marked to indicate pneumatic connections, fill and drain ports, and safety devices such as burst diaphragms and safety valves.

3.7.19 Parts list. The parts list for the starter that successfully completes the preproduction test shall constitute the approved parts list for subsequent starters of the same model (see 6.2). No changes shall be made in the design of parts or materials listed in the approved starter parts list except when such changes are approved in accordance with the provisions of MIL-STD-480.

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3.7.19.1 Material substitutions. Material substitutions or changes shall not be made without prior approval of the procuring activity or its authorized representative.

3.7.19.2 Approval of changes. Approval of changes does not relieve the contractor of full responsibility for the results of such changes on starter characteristics.

3.7.20 Changes in vendors. No changes shall be made in vendors or fabrication sources except when such changes are approved by the procuring activity.

3.7.21 Inspection processes and certification

3.7.21.1 Magnetic particle inspection. Magnetic particle inspection shall be in accordance with MIL-I-6868.

3.7.21.2 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be in accordance with MIL-I-6866.

3.7.21.3 Radiographic inspection. Radiographic inspection shall be in accordance with MIL-STD-453.

3.7.21.4 Hydrostatic test methods. Hydrostatic test methods shall be approved by the procuring activity.

3.7.21.5 Fusion welding certification. All operators performing fusion welding shall be certified in accordance with MIL-T-5021.

3.8 Performance characteristics. The starter performance characteristics shall be specified in the model specification and shall meet the starting and compatibility requirements of the engine(s) specified in the detail specification with which the starter is used. The starter shall operate within the limits specified in the model specification using a cartridge or a pneumatic energy source. All performance characteristics shall be based upon use of the production hardware.

3.8.1 Starter operation

3.8.1.1 Duty cycles. The characteristics of the starter shall permit the following duty cycles and operation within the limits specified in the model specification throughout the ambient temperature range of -65° to +160°F:

a. Two cartridge cycles within 5 minutes in any 60-minute period

b. Three pneumatic cycles in any 15-minute period

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c. In the event cartridge cycles and pneumatic cycles are interspersed, the total number of cycles shall be limited to three in any 15-minute period except that the condition specified in 3.8.1.1a is also applicable.

3.8.1.2 Altitude. The starter shall operate satisfactorily throughout the altitude range specified in the detail specification. The change in torque with altitude and the maximum altitude for satisfactory starter operation shall be specified in the model specification. In addition, the starter shall satisfactorily withstand overrunning conditions to maximum aircraft operational altitude.

3.8.1.3 Temperature

3.8.1.3.1 Cartridge cycle. The starter shall operate within the limits specified in the detail specification and suffer no detrimental effects after conditioning to -65°F, +160°F, or any intermediate temperature when using cartridges conditioned to -65°F, +160°F, or any intermediate temperature.

3.8.1.3.2 Pneumatic cycle. The starter, when conditioned to a temperature of -65°F, +160°F, or any intermediate temperature, shall operate within the limits specified in the detail specification at the inlet pressures, temperatures, and flows specified in the detail specification. The inlet conditions shall be specified in the model specification.

3.8.1.3.3 Overrunning. The starter shall be capable of overrunning continuously at the temperatures specified in the detail specification.

3.8.1.4 No-load operation. The starter shall be designed to operate throughout the temperature range specified in the model specification without structural failure when cartridge or pneumatic cycled with no load on the output shaft.

3.8.1.5 Endurance. The starter shall operate within the limits specified in the detail and model specifications throughout the overhaul life, and within the following performance deterioration limits:

a. Cartridge performance at the end of 400 cartridge cycles shall not deteriorate more than 5 percent from the values obtained during initial calibration.

b. Pneumatic performance at the end of 400 cartridge and 600 pneumatic cycles shall not deteriorate for any reason more than 5 percent from the performance obtained during initial calibration.

3.8.1.6 Environmental conditions. The starter shall suffer no detrimental effects and shall operate satisfactorily during and after exposure to the natural and induced environments including explosive atmosphere, salt fog, dust, fungus, humidity, and other environmental conditions as anticipated in service.

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3.8.1.6.1 Field operation. The starter shall operate within the limits specified in the model specification under field operating conditions.

3.8.1.7 Running engagement. The starter shall suffer no detrimental effects when subjected to running engagements.

3.8.2 Starter ratings. The starter sea level performance ratings shall be specified in the model specification.

3.8.3 Performance curves. The starter performance shall be defined by the following curves which shall be part of the model specification. The curves shall include minimum and maximum values and the performance limits specified in the model specification.

3.8.3.1 Curves shall be included in the model specification to show starter output torque versus rpm with starter and cartridges conditioned to -65°F, +59°F, and +160°F.

3.8.3.2 Curves shall be included in the model specification to show starter output torque versus rpm at -65°F, +59°F, and +160°F under the inlet air conditions from matching the air supply characteristics at the starter inlet. The curves shall show the maximum and minimum opening time of any valve, if applicable.

3.8.3.3 Curves shall be included in the model specification to show starter rpm versus time during operation with no load on the output shaft.

3.8.3.4 Curves shall be included in the model specification to show the variation of starter output torque (cartridge and pneumatic) versus time, torque delay time, torque rise time, starter cutoff speed, and operating time for -65°F, +59°F, and +160°F cartridge operation.

3.8.3.5 Curves shall be included in the model specification to show typical starter output torque impact with temperature for cartridge and pneumatic operation.

3.8.3.6 Curves shall be included in the model specification to show the variation with altitude of maximum output torque and torque delay time with temperature at sea level.

3.8.4 Performance variation. The greatest values of maximum impact torque, maximum torque, and maximum breech pressure that occur in the temperature range of -65° to +160°F shall be specified in the model specification (specify mode of operation). The contractor shall furnish curves showing starter output torque versus rpm at -65°F, +59°F, and +160°F (cartridge) and at rated, maximum temperature, minimum pressure, and maximum pressure pneumatic inlet conditions superimposed on engine starting characteristics specified in the detail specification.

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3.9 Details of components

3.9.1 Breech and gas transfer assembly. The breech and gas transfer assembly shall consist of those components required or necessary to transfer the cartridge or pneumatic energy to the energy conversion assembly.

3.9.1.1 Breech. The breech shall be so designed that it can be separated into two sections to facilitate cartridge insertion and removal and shall be such that the separable parts can be assembled to achieve closure and electrical continuity. The breech closure operation shall require no special or unusual techniques nor shall the use of any tools be required to effect the seal and closure.

3.9.1.1.1 Breech pressures. The minimum proof and burst pressures shall be 3,000 and 4,000 pounds per square inch (psi), respectively, when the breech is closed and locked. The actual minimum proof, minimum yield, and minimum burst pressures shall be specified in the model specification. The proof pressure shall be not greater than the minimum yield pressure.

3.9.1.1.2 Arming device. Any device utilized in arming the cartridge shall be described in detail in the model specification. The total direct-current (dc) resistance in the cartridge firing circuit shall not exceed .5 ohm excluding the resistance of the cartridge squib. The actual value shall be specified in the model specification.

3.9.1.2 Cartridge. The cartridge used shall be in accordance with MIL-C-27505 or MIL-C-27658. The cartridge type and model number or the cartridge model specification number shall be referenced in the starter model specification. Only Air Force approved cartridges shall be used. Approved cartridge vendor sources shall be coordinated with the procuring activity.

3.9.1.3 Nozzles. The cartridge and pneumatic nozzles shall be designed to minimize erosion and deposits during cartridge and pneumatic operation. The nozzle characteristics, flow coefficients, size(s), including production tolerances shall be specified in the model specification and be compatible with the cartridge and pneumatic energy sources specified in the model specification.

3.9.1.4 Overpressure device. An overpressure device which will limit the maximum breech pressure in the event of abnormal cartridge burning shall be incorporated into the starter. The device shall provide a path for excess cartridge gas which will bypass the normal flow path through the starter. This excess cartridge gas shall be channeled into the starter exhaust system. The external surface temperature of the starter shall not exceed that specified in 3.7.2.

3.9.1.4.1 Actuation pressure. The actuation pressure (with tolerances) of the overpressure device shall be specified in the model specification. The

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overpressure device shall actuate at a pressure less than 500 psi above the maximum operating pressure as applicable for satisfactory operation of the unit. The device shall function within the limits specified in the model specification throughout the operating life of the starter.

3.9.1.4.2 Exit area. The area of the opening through which the excess gas will flow after actuation of the overpressure device shall be specified in the model specification.

3.9.1.4.3 Replacement and operation. If the overpressure device is not automatically resettable, it shall be such that it can be easily inspected for actuation. Any overpressure device shall operate satisfactorily for a minimum of 400 cartridge cycles and shall be replaceable without the use of special tools and without removal of the starter from the aircraft. This device shall be described in the model specification.

3.9.1.4.4 Damage. Operation of the overpressure device shall not be detrimental to the starter or aircraft.

3.9.1.4.5 Gas leakage. The requirements of 3.7.4 shall apply upon operation of the overpressure device.

3.9.1.4.6 Indicator. Unless otherwise provided with an automatic reset feature, the overpressure device shall incorporate a mechanism which will provide visual indication that an overpressure condition has occurred.

3.9.1.4.7 Pneumatic operation. It shall not be necessary to replace or reset the overpressure device after it has operated to achieve a satisfactory pneumatic start.

3.9.1.5 Valves

3.9.1.5.1 Pneumatic control valves. The pneumatic control valve shall not be a part of the starter and shall not be furnished with the starter unless specified by the procuring activity. The starter shall be designed to operate satisfactorily with either a pressure-regulating or a nonregulating (on-off) starter control valve in accordance with MIL-V-38398. The pressure-regulating starter control valve will be used in installations where the pneumatic supply pressure exceeds 62 pounds per square inch absolute (psia). The nonregulating (on-off) starter control valve will be used in installations where the maximum pneumatic supply pressure is 62 psia or less. The type valve to be used will be specified in the detail specification. An automatic control for starter cutoff shall be provided as an integral part of the starter and shall be compatible with either type valve.

3.9.1.5.2 Cartridge gas pressure regulator. Any cartridge gas pressure regulator valve used to control starter gas flow in the gas transfer assembly

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shall operate within the limits specified in the model specification without service or adjustment for a minimum of 400 cartridge cycles. The maximum and minimum flow rates and tolerance variation (at specified pressures and temperatures) through the valve shall be specified in the model specification.

3.9.1.6 Pneumatic air source. Pneumatic air will be supplied by ground support equipment, engine innerbleed, or both, and the limits will be specified in the detail specification.

3.9.2 Energy conversion assembly. The energy conversion assembly converts the gas energy supplied by the cartridge or pneumatic energy source into mechanical energy.

3.9.2.1 Turbine

3.9.2.1.1 Polar moment of inertia. The polar moment of inertia of the complete turbine rotor about the rotor axis shall be specified in the model specification.

3.9.2.1.2 Operational and design speeds

3.9.2.1.2.1 Maximum free-running speeds. The turbine maximum free-running speeds for cartridge operation at -65°F, +59°F, and +160°F and pneumatic operation shall be specified in the model specification and shall not exceed 75 percent of the turbine minimum burst speed.

3.9.2.1.2.2 Minimum yield speed. The turbine minimum yield speed shall be greater than the maximum free-running speed and shall be specified in the model specification.

3.9.2.1.2.3 Turbine burst speed. The turbine minimum and maximum burst speeds shall be specified in the model specification.

3.9.2.1.2.4 Proof speed. The turbine proof speed shall be less than the minimum yield speed and greater than the maximum free-running speed and be specified in the model specification.

3.9.2.1.3 Turbine identification. Each turbine wheel shall be identified by serial number. The turbine wheel serial number, date of installation, and prior number of cycles shall be shown on a turbine identification plate. The identification plate shall be in accordance with MIL-STD-130. Sufficient space shall be provided on the plate to enter the appropriate turbine record through five overhaul periods.

3.9.2.2 Other means of energy conversion. Any other means of energy conversion shall be described in the model specification.

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3.9.3 Speed reduction assembly

3.9.3.1 Gears. The gear ratio between the energy conversion mechanism and the starter output shaft shall be specified in the model specification. The amount of gear backlash shall be the minimum consistent with the application and shall be specified in the model specification.

3.9.4 Clutch and output assembly

3.9.4.1 Clutches

3.9.4.1.1 Engaging clutch. An engage-disengage clutch shall be provided which will automatically engage or disengage the starter from the engine under all engine starting conditions. The clutch shall operate within the limits specified in the model specification.

3.9.4.1.2 Slip clutch. In the event a slip clutch is used, it shall be set to slip within ±5 percent of the maximum breakaway torque value specified in the model specification.

3.9.4.2 Output shaft. The maximum starter output shaft torque and direction of rotation shall be defined in the detail specification, and the actual maximum torque shall be specified in the model specification.

3.9.4.2.1 Shear section. A shear section shall be provided as part of the starter output shaft. The maximum applied torque value at which shear will occur shall be specified in the detail specification. The maximum and minimum values at which the shaft will shear shall be specified in the model specification.

3.9.4.3 Mounting flange and drive. The starter mounting flange and drive spline shall conform to the detail specification requirement and shall mount on the corresponding engine accessory drive. The accessory drive type number shall be specified in the model specification.

3.9.5 Control system. The starter control system shall operate in such a manner that it shall not be necessary to adjust the starter controls for various modes of starter operation.

3.9.5.1 Automatic cycling. Acceleration and operation of the starter through disengagement from the load shall be automatically controlled by the starter control system during the cartridge mode. Actuation of the cartridge cycle shall be controlled by a single momentary switch in the cockpit of the aircraft. Provisions in the starter for actuation or shutoff of the pneumatic source shall be required.

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3.9.6 Duct system. All ducting shall be of metal and shall incorporate quick disconnect features. In the event the starter requires secondary air from the nacelle or engine compartment, the air inlet shall be protected against ingestion of foreign objects.

3.9.6.1 Pneumatic inlet duct. The pneumatic inlet duct shall incorporate a mating device for the pneumatic starter control valve (see 3.9.1.5.1), and it shall be possible to fix the control valve on the duct to any angular position of the duct.

3.9.6.2 Exhaust duct. The exhaust system duct utilized shall have its dimensions and flows specified in the model specification and shall be compatible with the exhaust duct requirements of the aircraft installation as specified in the detail specification.

3.9.7 Power system

3.9.7.1 Electrical

3.9.7.1.1 Electrical power. The starter shall meet the requirements of this specification with electrical power from a 28 volt dc system conforming to MIL-STD-704. All electrical equipment shall be designed in accordance with MIL-E-25499.

3.9.7.1.2 Connectors and cable. Electrical connectors and cable shall be in accordance with MIL-C-5015 and MIL-W-25038. It shall be possible to connect and disconnect electrical connectors and to flex electrical conductors without damage throughout a temperature range of -65° to +160°F.

3.9.7.1.3 Cutoff switch. For pneumatic starts, the starter shall contain a speed-sensing device that will actuate an electrical switch of 1 ampere, 28-volt capacity to open the switch when the output shaft reaches the required engine assist speed as specified in the detail specification. This switch shall sense directly the speed of the output shaft. The switch must reclose at 1,400 rpm minimum.

3.9.7.1.4 Starter cartridge circuit. The maximum resistance through the starter cartridge circuit shall be as specified in the model specification and shall not exceed 0.5 ohm during or after the tests specified herein.

3.9.7.2 Other power systems. Any other type of power required by the starter shall be specified in the model specification and shall conform, where applicable, to military specifications.

3.9.8 Lubrication system. The lubricating system shall adequately lubricate the starter throughout its operating range. The starter design shall be such

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that no change in lubricant shall be required for operation throughout the ambient temperature range specified in the detail specification. It shall be possible for the starter to complete 500 cycles (cartridge and pneumatic) including 200 hours of overrunning at any temperature within the temperature range specified herein without adding to or changing the lubricant. Starter operating temperature range for cycling shall be -65° to +160°F. Overrunning temperature condition shall be as specified in the detail specification.

3.9.8.1 Oil supply. The oil reservoir shall be furnished as a component part of the starter. The consumption rate shall be specified in the model specification. It shall not be necessary to disassemble the starter in order to refill the reservoir. The lubricant used shall be in accordance with MIL-L-7808. The minimum amount of oil necessary for satisfactory starter operation shall be specified in the model specification.

3.9.8.2 Oil loss. Oil loss from the starter shall be such that it will not be ingested by the engine. The starter maximum oil loss rate per cycle shall be specified in the model specification.

3.9.8.3 Oil flow and heat rejection. In the event pressure lubrication is utilized, the oil flow and heat rejection rates shall be specified in the model specification.

3.9.9 Other components. The description and performance characteristics of other starter components not specified herein shall be described in detail in the model specification.

3.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern changes in the manufacturer's part numbers. Physical interchangeability shall be in accordance with MIL-I-8500.

3.11 Weight of starter

3.11.1 With cartridge. The weight of the starter filled with oil including the cartridge shall be specified in the model specification.

3.11.2 Without cartridge. The weight of the starter filled with oil without the cartridge shall be specified in the model specification and shall not exceed the maximum weight specified in the detail specification.

3.12 Finishes. All aluminum parts shall be anodized in accordance with MIL-A-8625. Unless otherwise specified by the procuring activity, all exposed surfaces except aluminum parts, corrosion-resisting steel, working surfaces, threads, and plated parts shall be finished with one coat of primer followed by two coats of black enamel.

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3.13 Drawings and diagrams. When specified (see 6.2), the contractor shall prepare the following drawings and diagrams for submission to the procuring activity with the model specification. In addition, reduced size copies of the same drawings and diagrams shall be incorporated in the body of the model specification.

a. Starter installation drawing, showing clearance required for removal, periodic inspections, identification, location, and definition of all connecting points

b. Electrical installation connection diagrams showing the circuits up to the connection points of the aircraft

c. A complete cutaway or cross-sectional drawing showing the starter's components and assembly.

3.14 Identification of product. The starter shall be marked for identification in accordance with MIL-STD-130 as follows:

Nomenclature
Manufacturer's Part No.
Manufacturer's Serial No.
Federal Stock No.
Design Activity Identification
Contract No.
Rotation (Output Shaft) 1/

1/ The arrow showing the direction of rotation may be included on a single nameplate with other specified data or, at the option of the contractor, it may be stamped on a separate plate attached immediately adjacent to the main nameplate.

3.15 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation, reliability, and durability consistent with the application and storage life requirements of the starter.

3.16 Deviations. The contractor shall include in the model specification under this paragraph a complete listing of all deviations to the requirements of this specification and the detail specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 All tests shall be subject to witnessing by an authorized representative or representatives of the procuring activity.

4.2 Classification of tests. The inspection and testing of starters shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions. Unless otherwise specified, the following conditions shall apply for all tests.

4.3.1 Temperature

4.3.1.1 Ambient temperature. Ambient temperature tests shall be conducted at the conditions existing within the test cell at the time of test.

4.3.1.2 Low temperature. Low temperature tests shall be conducted at -65°F.

4.3.1.3 Normal temperature. Normal temperature tests shall be conducted at 59°F.

4.3.1.4 High temperature. High temperature tests shall be conducted at 160°F with the exception of the overrunning tests as specified in 4.6.7.1.

4.3.2 Pressure. Unless otherwise specified herein, all tests shall be conducted at ambient atmospheric pressure.

4.3.3 Inlet conditions. Unless otherwise specified, all pneumatic cycling tests shall be conducted at rated inlet conditions. The pressure, temperature, and flow parameters for the following inlet conditions shall be specified in the detail specification for the applicable starter installation:

a. Rated - 45 psia, 450°F, and 90 pounds per minute

b. Maximum temperature

c. Minimum pressure

d. Maximum pressure.

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4.3.4 Conditioning time. Conditioning time for the starter or cartridge shall be such that all parts shall have reached a temperature within ±5°F of the specified temperature. A starter shall be considered conditioned when it has been continuously exposed to the specified temperature for the conditioning time, making suitable allowance for the starting temperature. During the conditioning time, the temperature of the conditioning chamber shall not vary more than ±5°F from the specified temperature.

4.3.4.1 Long term low temperature soak. The starter shall be maintained at the low temperature for a period of 72 hours.

4.3.5 Transfer time. The starter and component transfer time shall not exceed 5 minutes.

4.3.6 Test apparatus

4.3.6.1 Accuracy. For all starter and component tests except MIL-STD-810 environmental tests, the test apparatus shall be such as to insure that reported data will have a steady-state accuracy within ±2 percent of the maximum rated value, except that temperature shall be accurate within ±2°F and weight shall be accurate within ±0.2 percent. All apparatus shall be calibrated often enough to insure that this degree of accuracy is maintained. Calibration records shall be retained by the testing agency for 2 years and furnished to the procuring activity upon request.

4.3.6.2 Automatic recording equipment. Automatic recording equipment shall be used to obtain data during those parts of starter and component tests requiring evaluation of time versus starter or component variables.

4.3.6.3 Test flywheels. Two flywheels shall be used in the preproduction test. They shall have different moments of inertia as described below and shall be referred to as flywheel A and flywheel B.

4.3.6.3.1 Flywheel A. Flywheel A shall have a polar moment of inertia which will subject the starter to a loading condition similar to a normal engine starting cycle. This polar moment of inertia shall be defined in the detail specification and shall be specified in the model specification.

4.3.6.3.2 Flywheel B. Flywheel B shall have a polar moment of inertia which will subject the starter to a loading condition similar to an engine exhibiting early light off and rapid acceleration after light off. This polar moment of inertia shall be defined in the detail specification and shall be specified in the model specification.

4.3.6.4 Starter attitude during testing. All starter tests shall be conducted with the starter mounted in the most critical operational attitude. The mounting attitude shall be described in the model specification.

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4.3.7 Dimensional units. All dimensions shall be expressed in the English gravitational system of units.

4.3.8 Tests and inspections for preproduction and quality conformance approval. Tests and inspections required of the starter and its components for the purpose of establishing preproduction approval and quality conformance approval will be limited to those tests specified herein except as modified for a particular starter design. This limitation of tests and inspections does not relieve the contractor of responsibility for fulfilling all requirements of this specification.

4.4 Preproduction testing

4.4.1 Preproduction test method. When specified (see 6.2), a preproduction test method shall be prepared. The test method shall have been approved by the procuring activity prior to the initiation of preproduction testing. This document shall not be a part of the model specification; however, any deviation to MIL-S-27266 shall be specified in the model specification.

4.4.2 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6 and shall be conducted in the order listed therein except for the component tests.

4.4.3 Preproduction test data

4.4.3.1 General test data requirements

4.4.3.1.1 Temperature corrections. Performance characteristics shall not be corrected for ambient air temperature.

4.4.3.1.2 Pressure corrections. Performance characteristics shall not be corrected for ambient pressure.

4.4.3.2 Preproduction data summary. A summary shall be prepared by the contractor of all preproduction testing accomplished. It shall include, as a minimum, the general data required in the following paragraphs, unless otherwise specified in the detail specification, and all specific data required in the particular test paragraphs. Where the length of traces makes their inclusion difficult, only the initial and final portions and any other portions of significant interest shall be included. Component test reports covering all component tests conducted on each type of component shall also be prepared. All test reports shall be prepared in accordance with MIL-STD-831. (See 6.2.)

4.4.3.3 Limits. Performance limits shall be superimposed on all curves. These limits shall be those specified in the model specification.

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4.4.3.4 Cycle data. A preproduction data summary sheet shall be prepared for each preproduction test unit and shall include the requirements of figure 1 as a minimum. In addition, the following is required:

a. High speed trace of impact torque versus time with point of starter switch actuation shown

b. On the same recording sheet, a trace of acceleration versus time and torque versus time with point of starter switch actuation shown.

4.4.3.5 Post cycle data. The following minimum data shall be taken immediately after operation:

a. Maximum external surface temperature at cycle termination shall be recorded for all cartridge calibrations, rapid cycles, burst diaphragm tests, speed-limiting tests, cartridge-operated containment tests, and every fifth cartridge endurance cycle.

4.4.3.6 Reduced data. The following data shall be recorded or calculated:

a. Maximum breech pressure

b. Maximum impact torque

c. Average torque

d. Air consumption (lb/sec)

e. Airflow time into inlet

f. Time to cutoff speed (specify flywheel used)

g. Cutoff speed

h. Torque output at 1/3 and 2/3 specification cutout speed

i. Cartridge burn time, average chamber pressure, and 0.25 second pressure.

4.4.3.7 Disposition of preproduction test data. Preproduction test data shall be retained by the testing agency for 2 years and furnished to the procuring activity or authorized representative upon request.

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TEST DATA SUMMARY SHEET

Items under section 1 shall be recorded for all tests. Items under sections 2 and 3 shall be recorded for cartridge and pneumatic mode of operation respectively.

1. a. Date
   b. Serial number of starter
   c. Model number of starter
   d. Size of flywheel (slug ft^2)
   e. Ambient temperature (°F)
   f. Oil temperature (°F)
   g. Barometric pressure (in. Hg abs)
   h. Cutoff switch actuation speed (rpm)
   i. Actual dry weight of unit
   j. Maximum unit vibration

2. a. Cycle time (zero rpm to maximum assist speed) (seconds)
   b. Continuous record of chamber pressure (psig)
   c. Initial cartridge temperature (°F)
   d. Continuous record of torque (zero rpm to maximum assist speed) (lb-ft)
   e. Continuous record of output shaft speed (rpm)
   f. Assist speed (rpm)
   g. Cartridge manufacturer
   h. Cartridge lot number and serial number
   i. Chamber pressure and starter output torque at 1,000 rpm and 2,000 rpm

3. a. Acceleration time (zero rpm to 2,000 rpm) (seconds)
   b. Inlet air total pressure (psia)
   c. Inlet air total temperature (°F)
   d. Stall torque (lb-ft)
   e. Stall airflow (lb per minute)
   f. Continuous record of torque (zero rpm to maximum assist speed) (lb-ft)
   g. Continuous record of output shaft speed (rpm)
   h. No-load speed (rpm)
   i. Torque at starter output shaft speeds of 1,000 rpm and 2,000 rpm

FIGURE 1. Test Data Summary Sheet

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4.4.3.8 Preproduction test data analysis. An analysis of all preproduction test data shall be conducted by the contractor and shall be included in the preproduction test report. Any unusual condition, malfunction, failure, or out-of-specification performance shall be explained in detail in the test report, and the procuring activity shall be notified in writing as soon as any of the above conditions are evident.

4.4.3.8.1 Recalibration. The point at which a significant difference occurs in any parameter shall be recorded and the starter shall be recalibrated. If this point occurs prior to the last cycle of testing, the test shall be stopped until the procuring activity is notified and gives approval to proceed.

4.4.4 Maintenance, adjustment, or replacement of components or parts. Maintenance, adjustment, or replacement of components or parts on preproduction test samples shall not be permitted after initial acceptance of the starters, except as necessary to convert from pneumatic to cartridge operation or vice versa. Disassembly of any part of the starter to any extent prior to the final teardown inspection, except for replacement of spent cartridges, shall not be permitted.

4.4.4.1 Lubrication. Lubrication shall be accomplished prior to the cycling and endurance testing. After initiation of the testing, addition of oil shall not be permitted. The oil may be changed after completion of 200 hours of overrunning.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 Individual tests. Each starter and each high speed component shall be subjected to the individual tests specified in 4.5.1.1 and 4.5.1.2 as a minimum. Additional tests may be required depending upon the particular starter design.

4.5.1.1 Individual tests for starters. Each starter shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1)

b. Initial cycling (see 4.6.2)

c. Overrunning test (see 4.6.3).

4.5.1.2 Individual tests for high speed components. Each high speed component shall be subjected to the tests described in 4.6.5. These component tests shall be completed prior to initiation of the individual starter tests specified in 4.5.1.1.

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4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Each starter of the first 10 production units, every tenth of the next 90, and every fiftieth production configuration starter thereafter shall be subjected to the teardown inspection described in 4.6.4.

4.5.2.2 Sampling plan B. The first and every fiftieth production unit thereafter shall be subjected to the starter cartridge circuit test described in 4.6.7.9.

4.5.3 Rejection and retest. If any starter fails to meet the requirements of the quality conformance tests, the starter shall be rejected. Starters which have been rejected may be reworked to correct the defects and resubmitted for acceptance. Before submission of the reworked starters, full particulars concerning previous rejection and the action taken to correct the defects found in the original starter shall be furnished to the procuring activity or its authorized representative.

4.5.4 Quality conformance test method. A quality conformance test method, including performance parameters, shall be included as a part of the model specification.

4.5.5 Quality conformance test data. Performance data shall be obtained from tests conducted on starters submitted for acceptance and shall not be corrected for ambient air temperature or ambient pressure.

4.5.5.1 Quality conformance data summary sheet. A quality conformance data summary sheet shall be prepared in accordance with the applicable portions of figure 1 for each starter.

4.6 Test methods

4.6.1 Examination of product. Each starter shall be subjected to a careful inspection to insure that all dimensional tolerances are within specification and drawing limits and that all parts are correctly assembled.

4.6.2 Initial cycling. Each starter shall be subjected to a minimum of four cycles, two of which shall be pneumatic at rated inlet conditions, and one cartridge cycle at -65°F and one at 160°F. Performance shall be within the limits specified in the model specification curves or the unit shall be rejected.

4.6.3 Overrunning test. The starter output shaft shall be driven for 10 minutes at a speed equivalent to the maximum engine speed at the starter mounting pad as specified in the detail specification. Failure of the starter to remain disengaged shall be cause for rejection. All starters submitted for inspection in accordance with 4.6.4 shall be subjected to a total of 30 minutes of overrunning at the specified speed before teardown.

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4.6.4 Teardown inspection. After completion of the initial cycling and overrunning tests, all preproduction starters and the starters tested in accordance with sampling plan A (4.5.2.1) shall be disassembled sufficiently to allow a detailed inspection of all working parts. The extent of the disassembly shall be at the option of the procuring activity or its authorized representative. If at any time the disassembly discloses internal deficiencies after the initial cycling and overrunning tests which can only be detected by disassembly, the original schedule for teardown inspection shall be re-initiated. Final cycling shall be accomplished and shall consist of one pneumatic cycle at rated inlet conditions and one cartridge cycle at ambient temperature.

4.6.5 High speed components. Prior to conducting any starter tests, the following shall be accomplished on all high speed components:

a. Subject each part to X-ray inspection in accordance with MIL-STD-453. Any cracks or occlusions that will adversely affect starter performance shall be cause for rejection.

b. Subject each part to fluorescent penetrant inspection in accordance with MIL-I-6866. Any cracks or occlusions that will adversely affect starter performance shall be cause for rejection.

c. Proof spin each turbine wheel for 1 minute at room temperature at the proof speed.

d. Upon completion of items a, b, and c above, subject each turbine wheel to X-ray inspection to determine the extent of deformity, structural damage, and growth of cracks or occlusions. Any deformity, structural damage, growth of cracks or occlusions, or any new cracks shall be cause for rejection.

4.6.6 Calibration. Two cycles of cartridge operation shall be completed at each of the following three temperatures of the cartridge: -65°F, +59°F, and +160°F. Two cycles of pneumatic operation shall be completed at each of the following conditions: Rated, maximum temperature, minimum pressure, and maximum pressure inlet conditions with pressure limits of ±1 psia and temperature limits of ±15°F. Individual test units shall be rejected if initial preproduction test unit calibrations do not meet or exceed the minimum performance specified in the model specification.

4.6.7 Starter cycling and endurance test. Including the initial calibration, the starter shall be subjected to 400 cartridge cycles, 600 pneumatic cycles, and endurance running of the starter drive in the overrunning condition as specified in the detail specification. The order of testing shall be as outlined in table I. Each cycle shall be conducted as follows:

a. The starter shall be operated such that the load will be accelerated to the cutoff speed of the starter.

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b. Upon completing the operation specified in 4.6.7a, the output shaft of the starter shall be driven in excess of 105 percent of the starter cutoff speed for a period of 1 minute.

TABLE I. Order of Cycling and Endurance Testing

Order	Test Table No.	Flywheel
1	II	A
2	II	B
3	III	A
4	III	B
5	IV	A
6	IV	B
7	V	A
8	V	B
9	VI	A
10	VI	B

4.6.7.1 Following each group of approximately 100 test cycles, conducted as specified in 4.6.7a and 4.6.7b, the starter output shaft shall be driven (overrunning) at a speed equal to the maximum speed of the engine accessory drive pad, ±100 rpm, for a period of time equal to one-tenth of the total endurance overrunning time. The surrounding air and drive pad temperatures shall be as specified in 3.8.1.3.3.

4.6.7.2 Cycling and endurance test conditions. The starter cycling and endurance tests shall be conducted as shown in tables II through VI. All tests conducted on flywheel A shall also be conducted on flywheel B as shown in table I, except as noted in the appropriate tables. The manufacturer of cartridges to be used in the tests in tables II through VI will be specified at the time of contract.

4.6.7.2.1 Cartridge endurance. Cartridge endurance cycles shall be conducted with cartridges and the test starter conditioned as specified in the applicable test tables. Cartridge cycles shall be intermixed with the pneumatic cycles.

4.6.7.2.2 Pneumatic endurance. Pneumatic endurance cycles shall be conducted utilizing the pneumatic inlet condition specified in the applicable test tables.

4.6.7.3 Vibration. The starter shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category (b), mounted without vibration isolators and mounted directly on the airplane engine.

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TABLE II. Category I Cycling and Endurance Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.6	Calibration 1/	Amb	High	2	—	2	—	—
4.6.6	Calibration 1/	Amb	Normal	2	—	—	2	—
4.6.6	Calibration 1/	Amb	Low	2	2	—	—	—
4.6.6	Calibration 1/	Amb	—	—	—	—	—	2
4.6.7.2.1	Cartridge endurance	High	High	3	—	—	—	—
4.6.7.2.1	Cartridge endurance	Amb	Amb	27	—	—	—	—
4.6.7.2.2	Pneumatic endurance	Amb	—	—	13	13	13	13
4.6.7.2.1	Cartridge endurance	High	Amb	1	—	—	—	—
4.6.7.2.2	Pneumatic endurance	Amb	—	—	1	—	—	—
4.6.7.7	Rapid cycling (Cartridge)	Amb	High	2	—	—	—	—
4.6.7.7	Rapid cycling (Pneumatic)	Amb	—	—	3	—	—	—
4.6.7.7	Rapid cycling (Cartridge–Pneumatic)	Amb	High	2	1	—	—	—
Overrunning to be performed as specified in 4.6.7.1								

1/ Calibration shall be performed on both flywheels A and B before conducting any other test cycles. Flywheel B may be used first for the calibration cycles.

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TABLE III. Category II Cycling and Endurance Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.7.2.1	Cartridge endurance	Low	Low	3	---	---	---	---
4.6.7.2.1	Cartridge endurance	Amb	Amb	27	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	13	15	13	13
4.6.7.2.1	Cartridge endurance	Amb	Low	1	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	---	---	1	---
4.6.7.7	Rapid cycling (Cartridge)	Amb	Low	2	---	---	---	---
4.6.7.7	Rapid cycling (Pneumatic)	Amb	---	---	---	---	3	---
4.6.7.7	Rapid cycling (Cartridge-Pneumatic)	Amb	Low	2	---	---	1	---
Overrunning to be performed as specified in 4.6.7.1.								

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TABLE IV. Category III Cycling and Endurance Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.7.2.1	Cartridge endurance	High	Low	3	---	---	---	---
4.6.7.2.1	Cartridge endurance	Amb	Amb	16	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	13	15	13	13
4.6.7.4	Speed limiting (Cartridge)	(See Test)	Low	1	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	---	---	1	---
4.6.7.7	Rapid cycling (Cartridge)	Amb	Low	2	---	---	---	---
4.6.7.7	Rapid cycling (Pneumatic)	Amb	---	---	---	---	3	---
4.6.7.7	Rapid cycling (Cartridge-Pneumatic)	Amb	Low	2	---	---	1	---
	TOTAL CYCLES TO THIS POINT (Both flywheels)			200	92	90	102	82
4.6.6	Calibration 1/	Amb	High	2	---	2	---	---
4.6.6	Calibration 1/	Amb	Normal	2	---	---	2	---
4.6.6	Calibration 1/	Amb	Low	2	2	---	---	---
4.6.6	Calibration 1/	Amb	---	---	---	---	---	2
Overrunning to be performed as specified in 4.6.7.1.								

1/ Calibration shall be performed on both Flywheels A and B after the 200-cycle point.

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TABLE V. Category IV Cycling and Endurance Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.7.2.1	Cartridge endurance	Low	High	3	---	---	---	---
4.6.7.2.1	Cartridge endurance	Amb	Amb	27	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	13	13	13	13
4.6.7.2.1	Cartridge endurance	Amb	High	1	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	1	---	---	---
4.6.7.7	Rapid cycling (Cartridge)	Amb	High	2	---	---	---	---
4.6.7.7	Rapid cycling (Pneumatic)	Amb	---	---	3	---	---	---
4.6.7.7	Rapid cycling (Cartridge-Pneumatic)	Amb	High	2	1	---	---	---
Overrunning to be performed as specified in 4.6.7.1.								

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TABLE VI. Category V Cycling and Endurance Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.7.2.1	Cartridge endurance	Amb	Amb	30	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	6	6	6	5
4.6.7.4	Speed limiting (Pneumatic)	(See test)	---	---	---	1	---	---
4.6.7.7	Rapid cycling (Cartridge)	Amb	Normal	2	---	---	---	---
4.6.7.7	Rapid cycling (Pneumatic)	Amb	---	---	---	3	---	---
4.6.7.7	Rapid cycling (Cartridge-Pneumatic)	Amb	Normal	2	---	1	---	---
4.6.6	Calibration 1/	Amb	High	2	---	2	---	---
4.6.6	Calibration 1/	Amb	Normal	2	---	---	2	---
4.6.6	Calibration 1/	Amb	Low	2	2	---	---	---
4.6.6	Calibration 1/	Amb	---	---	---	---	---	2
4.6.7.3	Vibration 2/	Amb	---	---	---	---	---	---
Overrunning to be performed as specified in 4.6.7.1.								
TOTAL CYCLES TO THIS POINT (Both flywheels)				362	148	146	148	126

1/ Calibration shall be performed on both flywheels A and B at this point before continuing with the tests.
2/ Vibration shall be performed with a fired cartridge in the breech.

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TABLE VI. Category V Cycling and Endurance Tests (Cont)

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.6	Calibration	Amb	Normal	2	---	2	---	---
4.6.7.2.1	Cartridge endurance	Amb	Amb	31	---	---	---	---
4.6.7.5	Running engagement	Amb	---	---	---	24	---	4
4.6.7.6	Gas leakage	Amb	Amb	2	---	---	---	---
4.6.6	Calibration	Amb	Normal	2	---	2	---	---
4.6.7.4	Speed limiting (Cartridge)	(See test)	High	1	---	---	---	---
Overrunning to be performed as specified in 4.6.7.1.								
	TOTAL TEST CYCLES TO THIS POINT			400	148 +	174 +	148 +	130 = 600
4.6.7.8	Emergency disengagement	Amb	High	1	---	---	---	---
4.6.7.9	Circuit resistance	---	---	---	---	---	---	---
4.6.7.10	Teardown inspection	---	---	---	---	---	---	---
4.6.7.13	Hydrostatic pressure	---	---	---	---	---	---	---
4.6.7.11	Overspeed failure	Amb	High	1	---	---	---	---

All cycles on this sheet to be conducted utilizing Flywheel A only.

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4.6.7.4 Speed limiting. The starter shall be subjected to both cartridge and pneumatic no-load cycles conducted as follows: The starter shall be mounted in a manner duplicating a typical installation, except that there shall be no load on the output shaft. The most critical environmental conditions which have an effect on the functioning of the speed-limiting feature shall be simulated. The starter shall be operated and the speed-limiting feature allowed to function normally during the entire duration of the cycle. Any evidence of deformity or structural damage shall be cause for rejection.

4.6.7.5 Running engagement. The flywheel shall be driven at engaging speed, and the starter engaged and accelerated to the maximum starter assist speed. Acceleration of the starter shall be controlled by actuation of the applicable starter control valve operating at the maximum pressure rise rate. Engaging speeds of the test shall be 500, 1,000, 1,500, and 2,000 rpm. This cycle shall be repeated six times (24 engagements) at the rated air inlet condition. The cycle shall then be repeated one time (four engagements) at the maximum engine inner bleed condition. Oscillograph traces shall be used for all engagements to record the following data:

a. Starter output torque (lb-ft)

b. Starter inlet total pressure (psig)

c. Starter inlet total temperature (°F)

d. Ambient temperature (°F)

e. Speed (rpm)

f. Electrical indication of start switch actuation.

4.6.7.6 Gas leakage. All preproduction test units shall be tested for gas leakage. The starter shall be installed in a chamber of approximately 10 ft^3. The starter exhaust shall be ducted to the outside, and any auxiliary air device on the starter shall be ducted to draw air from outside the compartment. A minimum of five air samples shall be taken within 2 inches of the starter at each of the following conditions. The samples shall be taken preceding a cartridge start, immediately after cartridge ignition, and near cartridge burnout. The test shall be conducted a minimum of two times with the starter and cartridge at ambient conditions. The air samples shall be analyzed and any concentration above 0.5 percent by volume of combustible gases shall be cause for rejection. The test procedure and air sample pickup location shall be approved by the procuring activity.

4.6.7.7 Rapid cycling. The starter shall be operated for eight cycles as follows:

a. Cartridge operate the starter for one cycle.

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b. Within a maximum of 5 minutes, again cartridge operate the starter for one cycle.

c. A maximum of 60 minutes after completing the operation specified in 4.6.7.7b, make three pneumatic cycles within 15 minutes.

d. A maximum of 60 minutes after completing the operation specified in 4.6.7.7c, make two cartridge cycles and one pneumatic cycle within 15 minutes.

4.6.7.8 Emergency disengagement. The starter shall be subjected to an emergency disengagement test conducted as follows:

a. After rendering the normal disengagement mechanism inoperative, the output shaft shall be driven until the emergency disengagement mechanism of the starter functions.

b. Upon completion of the operation specified in 4.6.7.8a and while the starter is in the emergency-disengaged position, cartridge operate the starter with a 160°F cartridge. The starter shall suffer no detrimental effects.

4.6.7.9 Starter cartridge circuit. All preproduction test units and the starters tested in accordance with sampling plan B (see 4.5.2.2) shall be subjected to a circuit test to determine the total direct-current resistance from pin "A" to pin "B" of the MS connector. The resistance shall not exceed the value specified in 3.9.1.1.2. The test shall be conducted using the cartridge specified in the detail specification with the squib terminal shorted to the cartridge case in such a manner as to not interfere with the normal method of grounding the cartridge to the breech. Resistance shall be measured by means of an instrument having an accuracy of 1 percent or better. The test shall be repeated five times for each cartridge model. Before each reading, the breech cover shall be removed, the cartridge removed, rotated, and reinserted, and the breech cover replaced. The maximum permissible resistance through the circuit of a new starter shall be determined during preproduction testing and shall be included in the model specification as the test parameter.

4.6.7.10 Endurance and cycling test teardown inspection. After completion of the required tests, the starters and components shall be disassembled for measurements, examination of parts, and photographs.

4.6.7.11 Overspeed failure. Upon completion of endurance testing, the starter shall be subjected to an overspeed condition, using the energy source specified in the applicable test table, such that a structural failure will occur within the starter. Any of the following conditions shall be cause for rejection:

a. Failure of the starter to remain on its mounts or of the housing to contain fragments. (It will be permissible for parts to fall from the starter exhaust provided the parts have zero kinetic energy.)

b. If cartridge operation is used to produce the failure, leakage of fire or combustion products at any area other than through the starter exhaust outlet.

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c. Surface temperature of the starter in excess of 700°F.

4.6.7.12 Hub containment. A starter shall be subjected to a turbine wheel hub containment at the maximum normal cutout speed. The turbine shall be suitably weakened to provide a tri-hub burst at the maximum normal cartridge assist speed. The conditions for rejection shall be as specified in 4.6.7.11.

4.6.7.13 Hydrostatic pressure. After all specified testing, the overpressure device shall be removed from the starter and subjected to a hydrostatic pressure test. The actuation or burst pressure shall be as specified in the model specification.

4.6.8 Starter environmental tests. Environmental testing shall include the following tests and shall be accomplished in the order specified in table VII. Unless otherwise specified herein or by the procuring activity, a starter other than the starter used in endurance testing shall be tested in the operational configuration. The starter tests shall be divided into groups as shown in table VII. The manufacturer of cartridges to be used will be specified at time of contract.

4.6.8.1 Altitude. The starter shall be conditioned and operated at a simulated altitude for engine starting specified in the detail specification, using conditioned cartridges and pneumatic inlet conditions as specified in table VII.

4.6.8.2 Humidity. The starter shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I. The starter shall be subjected to one cartridge and two pneumatic calibration cycles before and after the test and one cartridge and one pneumatic cycle at the end of 50, 100, 150, and 200 hours using conditioned cartridges and pneumatic inlet conditions as specified in table VII.

4.6.8.3 Salt fog. The starter shall be subjected to a 50-hour salt fog test in accordance with MIL-STD-810, method 509. The starter shall be subjected to one cartridge and two pneumatic calibration cycles before and after the test and one cartridge and one pneumatic cycle at the end of 5, 10, 20, and 30 hours using conditioned cartridges and pneumatic inlet conditions as specified in table VII.

4.6.8.4 Dust. The starter shall be subjected to a sand and dust test in accordance with MIL-STD-810, method 510. After being exposed to this test, the starter shall be operated using cartridges and pneumatic inlet conditions as specified in table VII.

4.6.8.5 Fungus. If the starter contains components of organic material, it shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. After being exposed to this test, the starter shall be operated using cartridges and pneumatic inlet conditions as specified in table VII.

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TABLE VII. Environmental Tests

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.6	Calibration	Amb	High	2	---	2	---	---
4.6.6	Calibration	Amb	Normal	2	---	---	2	---
4.6.6	Calibration	Amb	Low	2	2	---	---	---
4.6.6	Calibration	Amb	---	---	---	---	---	2
4.6.7.2.1	Cartridge endurance	Amb	Amb	50	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	30	30	30	30
4.3.1.4	High temp	High	High	6	8	---	---	---
4.3.1.3	Normal temp	Normal	Normal	6	8	---	---	---
4.3.1.2	Low temp	Low	Low	6	4	---	---	4
4.6.8.1	Altitude	Amb	Normal	6	2	2	2	2
4.6.8.1	Altitude	Amb	Low	6	2	2	2	2
4.6.7.2.1	Cartridge endurance	Amb	Amb	22	2	2	2	2
4.6.7.2.2	Pneumatic endurance	Amb	---	---	16	16	16	16
4.6.8.2	Humidity	Amb	Normal	6	---	8	---	---
4.6.8.3	Salt fog	Amb	Normal	6	---	8	---	---
4.6.8.4	Dust	Amb	Amb	6	2	2	2	2
4.6.8.5	Fungus	Amb	Amb	6	2	2	2	2
4.3.4.1	Temperature soak	Low	Low	1	---	---	---	---
4.6.8.6	Explosive atmosphere	Amb	Amb	3	---	---	---	---

Note: All tests shall be conducted on Flywheel A only.

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TABLE VII. Environmental Tests (Cont)

Test Paragraph No.	Title of Test	Starter Temp	Cartridge Conditioned Temp	Cartridge Cycles	Inlet Conditions for Pneumatic Cycles			
					Maximum Temp	Rated	Minimum Pressure	Maximum Pressure
4.6.8.7	Overpressure device test	High	High	2	---	---	---	---
4.6.8.7	Overpressure device test	Normal	Normal	2	---	---	---	---
4.6.8.7	Overpressure device test	Low	Low	2	---	---	---	---
4.6.7.6	Gas leakage	Amb	Amb	2	---	---	---	---
4.6.7.2.1	Cartridge endurance	Amb	Amb	50	---	---	---	---
4.6.7.2.2	Pneumatic endurance	Amb	---	---	30	30	30	30
4.6.7.9	Starter cartridge circuit	---	---	---	---	---	---	---
4.6.6	Calibration	Amb	High	2	---	2	---	---
4.6.6	Calibration	Amb	Normal	2	---	---	2	---
4.6.6	Calibration	Amb	Low	2	2	---	---	---
4.6.6	Calibration	Amb	---	---	---	---	---	2
4.6.7.10	Teardown inspection	---	---	---	---	---	---	---
4.6.7.13	Hydrostatic pressure	---	---	---	---	---	---	---
4.6.7.12	Hub containment	Amb	Amb	---	---	---	---	---

Note: All tests shall be conducted on Flywheel A only.

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4.6.8.6 Explosive atmosphere. The starter shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, method 511, procedure I. Three cartridge cycles shall be accomplished during the test with the explosive mixtures in the intermediate, lean, and rich sections of the explosive range. Any explosion external to the starter shall constitute failure.

4.6.8.7 Overpressure device test. This test shall be conducted five times as follows: Plug the starter nozzles sufficiently to simulate a stopped nozzle and then fire a cartridge. This should cause the burst diaphragm or valve to relieve the pressure buildup. Within a 5-minute time period, fire a second cartridge after the burst diaphragm has ruptured or the valve has actuated. There shall be no leakage or fire combustion products at any area other than through the starter exhaust, and the external surface temperature of the starter shall not exceed 700°F. Breech pressure and maximum surface temperatures shall be recorded throughout the test.

4.6.8.8 Additional tests. Additional starter tests for the purpose of demonstrating other special features of the starter may be required by the procuring activity. These tests shall be as specified in the detail and model specifications.

4.6.9 The following tests may be conducted on either the endurance or environmental test units after completion of that unit's particular tests (with the exception of the overspeed failure test) or on a separate unit or separate components, as applicable. These tests may be conducted in any sequence desired.

4.6.9.1 Previous component test approval. These tests may be waived on starter components requiring testing as specified herein at the option of the procuring activity, provided the component has been previously subjected to the tests specified herein and approved. With the exception of mounting provisions, such components must be substantially identical to the respective components previously approved. If such a waiver is granted, information on the components for which previous approval was obtained shall be provided as part of the preproduction test report.

4.6.9.2 Breech proof pressure. Five complete breech assemblies shall be subjected to a hydrostatic proof pressure test. Safety diaphragms or valves shall be sealed. The proof pressure shall be as specified in the model specification. There shall be no evidence of yield or other detrimental effects.

4.6.9.3 Overpressure device test. Nine complete overpressure devices shall be used for this test. Three each of the devices shall be conditioned to temperatures of -65°F, +700°F, and +2,000°F. Each device shall then be subjected to a gas pressure test such that actuation will occur in the case of a valve and burst will occur in the case of a diaphragm. Actuation or burst pressure(s) shall be as specified in the model specification.

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4.6.9.4 Clutch loading. The starter or complete clutch assembly shall be mounted on a locked brake and provisions made to rotate the driving side of the clutch a minimum of 90°. The driving side of the clutch shall be rotated a minimum of 90° for 50 times. The interval between these tests shall not cause the clutch to overheat. After completion of the test, the clutch loading shall not have changed more than ±15 percent from the nominal torque value specified in the model specification.

4.6.9.5 Starter - control valve compatibility. If the intended application for the starter requires the use of a starter control valve in the installation, the starter shall be subjected to a minimum of 100 test cycles with pneumatic inlet conditions to the starter equivalent to the maximum pressure rise rate of the valve at maximum regulated pressure condition or maximum engine interbleed pressure condition, as applicable. These cycles may be conducted in conjunction with the pneumatic endurance cycles required in 4.6.7.2.2 at the required inlet condition.

4.6.9.6 Turbine wheel burst. Two turbine wheels that have successfully passed the inspections and tests under 4.6.5 shall be subjected to burst tests as follows. Wheel speed and temperature at burst shall be recorded.

a. One turbine wheel shall be spun to burst speed at normal temperature.

b. One turbine wheel shall be spun to burst speed at maximum turbine temperature expected during operation.

4.6.9.7 Shear section. Five shear sections shall be operated for 2,000 cycles from zero to maximum torque under normal starter operating conditions without failure. Upon completion of the 2,000 cycles, the section shall be loaded to failure. The loads causing failure shall be as specified in the model specification. The rate of loading shall simulate the maximum rate of loading to which the starter is subjected during normal operation.

4.6.9.8 Load. The unit and its supports shall be subjected to a static load as specified in the detail specification acting through the center of gravity downward and in 20° increments to 80° either side of the bottom of the starter as mounted in the aircraft. A complete examination for structural failure of the unit shall be made at completion of the load test. This examination shall include X-ray inspection of the unit and mounting flange. Any evidence of deformity or structural damage shall be cause for rejection.

4.6.9.9 Other components (hydrostatic test). The following items shall be hydrostatically tested. Evidence of deformation adversely affecting starter operation or permanent set shall be cause for rejection.

a. The maximum starter operating pressure shall be imposed on the burst diaphragms or valves and held for a minimum period of 1 minute. Upon increasing

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the pressure, the burst diaphragm shall rupture or the valve shall open before reaching 500 psi above the maximum operating pressure.

b. All starter components subject to gas pressure during starter operation shall be tested for at least 20 seconds, either as separate units or collectively as subassemblies, to a pressure two times the maximum starter operating pressure. Suitable plugs or blanking plates may be used. These components and their proof and burst pressure shall be specified in the model specification.

4.6.9.10 Additional component tests. Additional components for the purpose of demonstrating other special component features may be required by the procuring activity. These tests shall be as specified in the model specification.

4.6.9.11 Teardown inspection. After completion of the required testing, the components and starters shall be disassembled for measurements, examination of parts, and photographs.

4.7 Inspection of preparation for delivery. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents referenced therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Preservation and packaging. Unless otherwise specified in the applicable contract or order, the oil used in preservation of the starter may conform to either MIL-L-7808 or MIL-C-8188. If MIL-C-8188 oil is to be used, the contractor may, at his discretion, conduct the last acceptance run using undiluted MIL-C-8188 oil as the starter lubricant, after which all excess oil shall be drained. The exposed mounting flange and drive shall be cleaned and coated with a light film of either MIL-L-7808 or MIL-C-8188 oil.

5.1.2 External preservation and packaging level A. After external preservation of the starter in accordance with 5.1.1, the starter, unless otherwise specified by the procuring activity, shall be preserved and packaged in accordance with method II, submethod IIa of MIL-P-116.

5.1.3 Level C. Starters shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

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5.2 Packing

5.2.1 Level A. Starters preserved and packaged as specified in 5.1.2, and unless otherwise specified by the procuring activity, shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Containers shall be of uniform shape and size and of minimum cube and tare weight consistent with the protection required.

5.2.2 Level B. Starters preserved and packaged as specified in 5.1.2, and unless otherwise specified by the procuring activity, shall be packed in domestic shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Unless otherwise specified by the procuring activity, starters shall be packed to afford protection against damage to ensure arrival at their destination in a satisfactory condition, and to be acceptable to the carrier at the lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules. This level may be the supplier's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be "Starter, Engine Cartridge/Pneumatic, Shaft Drive."

5.4 Packing list. The contractor shall furnish a packing list with each starter. All parts that are not installed on the starter, but are shipped with it, shall be included on the packing list.

6. NOTES

6.1 Intended use. The starters covered by this specification are intended for use in aircraft and missiles using gas turbine engines.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. Requirements for submission of the following data:

(1) Model specification (see 3.2)

(2) Drawings and data specified in 3.13

(3) Preproduction test method and test reports (see 4.4.1 and 4.4.3.2).

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c. The parts list should be delivered to the procuring activity prior to delivery of production hardware (see 3.7.19).

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Definitions and symbols. The applicable definitions and symbols used herein and in the model specification are as follows:

6.3.1 Procuring activity. The procuring activity is the Department of Defense activity which is responsible for the starter contract.

6.3.2 Rating. A rating is the value of some characteristic of performance as specified in the model specification.

6.3.3 Estimate. An estimate is a predicted value of performance.

6.3.4 Standard conditions. Standard conditions are the values of air temperature and pressure given in MIL-STD-210.

6.3.5 Burst pressure. Burst pressure is that applied internal pressure which causes the ultimate strength of a starter part to be exceeded.

6.3.6 Yield pressure. Yield pressure is that applied internal pressure which causes a starter part to be stressed to the yield point.

6.3.7 Proof pressure. Proof pressure is that internal pressure to which all high pressure starter parts may be subjected without causing structural integrity damage.

6.3.8 Initial yield. Initial yield is that point at which 0.2 percent permanent deformation has occurred.

6.3.9 Transfer time. Transfer time is the time interval from the removal of the starter or component from one conditioning chamber to the insertion of the starter or component into another conditioning chamber, or the time interval from the removal of the starter or component from the conditioning chamber to the initiation of testing.

6.3.10 Zero time. Zero time is that time at which starting switch starting actuation occurs.

6.3.11 Torque delay time. The torque delay time is the time from zero time to the point on the torque trace when the torque has risen to 10 percent of the maximum torque.

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6.3.12 Torque rise time. The torque rise time is the time required for the torque to rise from 10 percent of the maximum torque to 75 percent of the maximum torque.

6.3.13 Rapid cycle. A rapid cycle is the operation of the starter two or more times within less than 60 minutes as specified herein.

6.3.14 Free-run. Free-run is the operation of the starter with no-load on the output shaft of the starter.

6.3.15 Cutoff. Cutoff is the speed at which the starter ceases to drive the output shaft.

6.3.16 Overrunning. Overrunning is the driving of the starter output shaft by some means other than by energizing the starter turbine.

6.3.17 Overspeed. Overspeed is the rotation of high speed starter parts at a speed which causes the yield strength of the parts to be exceeded.

6.3.18 Aerodynamic speed limiting. Aerodynamic speed limiting is the prevention of the starter turbine from reaching minimum burst speed by some aerodynamic means that is inherent within the turbine design.

6.3.19 Impact torque. Impact torque is the maximum torque delivered to the output shaft of the starter, when engaged to the engine, due to gear backlash during initial operation.

6.3.20 Maximum torque. Maximum torque is the highest torque value occurring after the initial impact torque occurs.

6.3.21 Cartridge operating time. Cartridge-operating time is defined as the number of seconds from zero time until cartridge burnout.

6.3.22 Cartridge burn time. Cartridge burn time is the elapsed time between the point where the pressure has risen to 50 percent of the maximum chamber pressure and the point where the pressure has fallen to 50 percent of the maximum chamber pressure excluding the ignition peak.

6.3.23 Average breech pressure. The average breech pressure is equal to the area under the pressure time curve encompassed by action time and divided by burn time. Action time should be as specified in MIL-C-27505.

6.3.24 Model specification. The model specification referred to herein is defined as the starter vendor prepared document used to define a particular starter.

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6.3.25 Detail specification. The detail specification referred to herein is defined as the document used to define the applicable design and performance parameters required for a particular starter application.

6.3.26 Symbols

Symbols	Quantity	Unit
A	Area	$in.^2$
Ae	Exhaust nozzle exit area(s)	$in.^2$
g	Gravitational constant	ft/sec^2
t_d	Torque delay time	sec
t_r	Torque rise time	sec
T	Temperature	^oF
Ws	Total weight of the starter	lb
Ns	Output shaft speed	rpm
N_t	Turbine speed	rpm
P	Pressure	psi
τ	Torque	lb-ft
τ_{max}	Maximum torque	lb-ft
τ_i	Impact torque	lb-ft

6.4 Limitation of demonstrations. Demonstrations required of a starter and its components for the purpose of establishing a service type approval or production acceptance will be limited to those tests specified herein as modified by the individual starter detail and model specification. This limitation of demonstrations will not relieve the starter manufacturer of responsibility for fulfilling all model specification requirements.

6.5 Asterisks have not been used to indicate changes over the previous issue of the specification due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

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APPENDIX

STARTER, ENGINE, CARTRIDGE/PNEUMATIC, SHAFT DRIVE, MODEL SPECIFICATION FOR (OUTLINE AND INSTRUCTIONS FOR PREPARATION)

10. SCOPE

10.1 This appendix establishes the format to be used by manufacturers in the preparation of specifications for cartridge/pneumatic engine starters.

20. APPLICATION

20.1 A complete model specification shall be prepared in accordance with the outline and instructions for preparation as specified herein. For each specific starter model, new starter type designations will be assigned by the procuring activity. When a new type designation is assigned, a new starter model specification shall be prepared with a new specification number assigned thereto by the starter manufacturer. The appropriate approval status shall be shown on page 1 of the model specification. No changes to a starter model specification shall be submitted to the procuring activity by means of amendments prior to the time when the model specification for the proposed starter type specified is approved and becomes a part of a contract. Changes in amendment form to an approved starter model specification that has been released and that forms a part of a contract shall be submitted to the procuring activity for approval. A model specification may be revised only to incorporate previously approved amendments.

20.2 The headings and numbering of sections and paragraphs in the model specification shall correspond to those of the general specification. Specific data required by the model specification, the detail specification, and such other data as may be required to completely describe the unit shall be included in the model specification. Omission of reference in the model specification to a particular requirement of the general specification shall be interpreted as compliance therewith. When departures from the requirements of this specification are necessary, the details of such departures shall be stated as a specific requirements bearing the same section, paragraph heading, and numbering as in this specification.

20.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the starter manufacturer for insertion of proper information and data related thereto in connection with the preparation of the model specification. Parenthetical statements shall not be copied in the model specification.

20.4 The specification number shall be the number assigned by the starter manufacturer. When a revised model specification is submitted, it shall be designated by the use of a dash and a letter following the number, with a revision date therefor, which shall be shown on page 1 only. Only the specification number and the revision suffix letter, if applicable, shall be shown on subsequent pages.

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20.5 For purposes of permitting preliminary evaluation of a proposed starter design, or for release of approved starter performance characteristics in connection with a design competition, the starter manufacturer shall submit a preliminary model specification to serve until superseded by a complete coordinated model specification which will be required for a production contract.

20.5.1 The preliminary model specification shall be prepared in accordance with the requirements stipulated herein.

20.6 All curves shall be presented on graph paper having an adequate number of subdivisions to permit easy interpolation.

30. MODEL SPECIFICATION

30.1 The format and content for the model specification shall conform to the following.

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(Number and title. The number and title shall be as follows:)

Approval Status (Include applicable statement) (Spec. No.)_____

(This draft has not been approved. DO NOT USE FOR PROCUREMENT PURPOSES)
(Approved by using service letter_____)

(Date)_____
(a)Revised_____(Date)
(b)Revised_____(Date)

MODEL SPECIFICATION

STARTER, ENGINE, CARTRIDGE/PNEUMATIC, SHAFT DRIVE,
(Insert service type and model designation if assigned)

(NAME OF CONTRACTOR)

(CONTRACTOR'S MODEL DESIGNATION)

1. SCOPE

1.1 Scope. This specification covers the requirements for the_____ (insert the service type and model designation) starter.

1.2 Classification. The_____(insert type and model) starter is a _____(insert a brief description of the salient features of this model starter).

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification. (List all military documents referenced in the basic specification. This listing shall include the specification number, revision, amendment number, date, and the complete specification title. Example: MIL-Z-0000B-1, 4 November 1958, Zippers, Nonlocking, General Specification for.)

2.2 The following nonmilitary documents (of the issue specified) form a part of this specification. (List all nonmilitary documents referenced in the basic specification and in the detailed model specification.)

3. REQUIREMENTS

3.2 Model specification. The headings and numbering of sections and paragraphs in this model specification correspond to those of MIL-S-27266 and the detail specification_____. Omission of reference in this specification

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to a particular requirement of MIL-S-27266 or the detail specification_____ shall be interpreted as compliance therewith.

3.7 Design and construction. The starter shall be designed for operational use with a cartridge and pneumatic energy source. The starter shall be compatible with cartridges designed and constructed in accordance with_____ and pneumatic energy sources available from_____.

3.7.1.1 Auxiliary air. The rate of airflow required by (name application) at temperatures of -65°F, +59°F, and +160°F is _____lb/sec, _____lb/sec, and _____lb/sec, respectively. The duct diameter is _____±_____ inches. The length of the duct required is _____± _____inches.

3.7.5 Interlock prevention. The point of actuation of the gearbox switch is _____± _____ rpm. The point of actuation of the turbine lockout switch is _____± _____rpm. The drift between the two settings is plus _____ rpm or _____ rpm per 10 cycles of starter operation.

3.7.6 Emergency disengagement. (Give a detailed description including operating limits of the emergency disengagement device.) It is set to operate at _____ ± _____ rpm.

3.7.7 Flight and ground loading conditions. The design limit torque load shall be _____ lb/ft, and the maximum torque loading conditions resulting from turbine wheel seizure during containment is _____ lb-ft.

3.7.8 Overhung moment. The maximum overhung moment of the starter, filled with oil, including an unfired cartridge, is _____ lb-in. The starter center of gravity locations, weights, and overhung moments are specified on outline drawing number_____.

3.7.9 Polar moment of inertia. The starter effective mass polar moment of inertia at the starter drive is _____ slug ft^2.

3.7.11 Maintainability. The following list includes all actions and time requirements for field maintenance of the starter:

Action	Time (Man-Minutes)
_____ flight hour oil check	_____ man-minutes
200 flight hour oil change	_____ man-minutes

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3.8 Performance characteristics. All performance characteristics are based upon use of the production hardware. The starter operating envelopes shall be as shown on figure_____. (This figure shall be drawn in accordance with figure 1.) The starter shall perform within the envelope of performance defined by tables I and II between the temperature limits of -65°F and +160°F. (Specify transfer time for starter and cartridge respectively.)

3.8.2 Starter ratings. The starter performance ratings at standard sea level static conditions shall be as listed in tables I and II.

3.8.3 Performance curves. Performance shown on figures_____ to _____ inclusive, constitute part of this specification. These curves indicate performance attained under specified conditions. (The following performance curves, prepared in accordance with figures 2 through 12 and the instructions specified herein, shall illustrate the performance obtainable. Where applicable, dotted curves shall be superimposed on each curve showing the tolerance on performance.)

3.8.3.1 Typical sea level starter output torque vs rpm when utilizing cartridges is shown on the following figures:

Starter and cartridge at -65°F	Figure _____
Starter and cartridge at +59°F	Figure _____
Starter and cartridge at +160°F	Figure _____

(These curves shall be drawn in accordance with figure 2.)

3.8.3.2 Typical sea level starter output torque vs rpm when utilizing a pneumatic energy source is shown on the following figures:

Starter at -65°F, inlet conditions at maximum	Figure _____
Starter at +59°F, inlet conditions at rated	Figure _____
Starter at +160°F, inlet conditions at minimum	Figure _____

(These curves shall be drawn in accordance with figure 2.)

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TABLE I. Performance Ratings at Standard Sea Level Static Conditions
(Cartridge Operated)

Parameter	Cartridge Conditioned Temp °F	AVERAGE Rating	Limits			
			Magnitude		Percent 1/	
			Max	Min	Max	Min
Maximum Output Torque	− 65 + 59 + 160					
Impact Torque	− 65 + 59 + 160					
Average Torque	− 65 + 59 + 160					
Maximum RPM at No-Load	− 65 + 59 + 160					
Cutoff Speed	− 65 + 59 + 160					

Parameter	Cartridge Conditioned Temp °F	AVERAGE Rating	Limits			
			Magnitude		Percent 1/	
			Max	Min	Max	Min
Operating Time	−65 +59 +160					
Delay Time	−65 +59 +160					
Rise Time	−65 +59 +160					
Maximum Breech Pressure	−65 +59 +160					
Average Breech Pressure	−65 +59 +160					

1/ Based on AVERAGE rating.

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TABLE II. Performance Ratings at Standard Sea Level Static Conditions (Pneumatic Operated)

Parameter	Inlet Conditions	AVERAGE Rating	Limits			
			Magnitude		Percent 1/	
			Max	Min	Max	Min
Maximum Output Torque	Maximum Rated Minimum					
Impact Torque	Maximum Rated Minimum					
Average Torque	Maximum Rated Minimum					
Maximum RPM at No-Load	Maximum Rated Minimum					
Cutoff Speed	Maximum Rated Minimum					
Operating Time	Maximum Rated Minimum					

1/ Based on AVERAGE rating.

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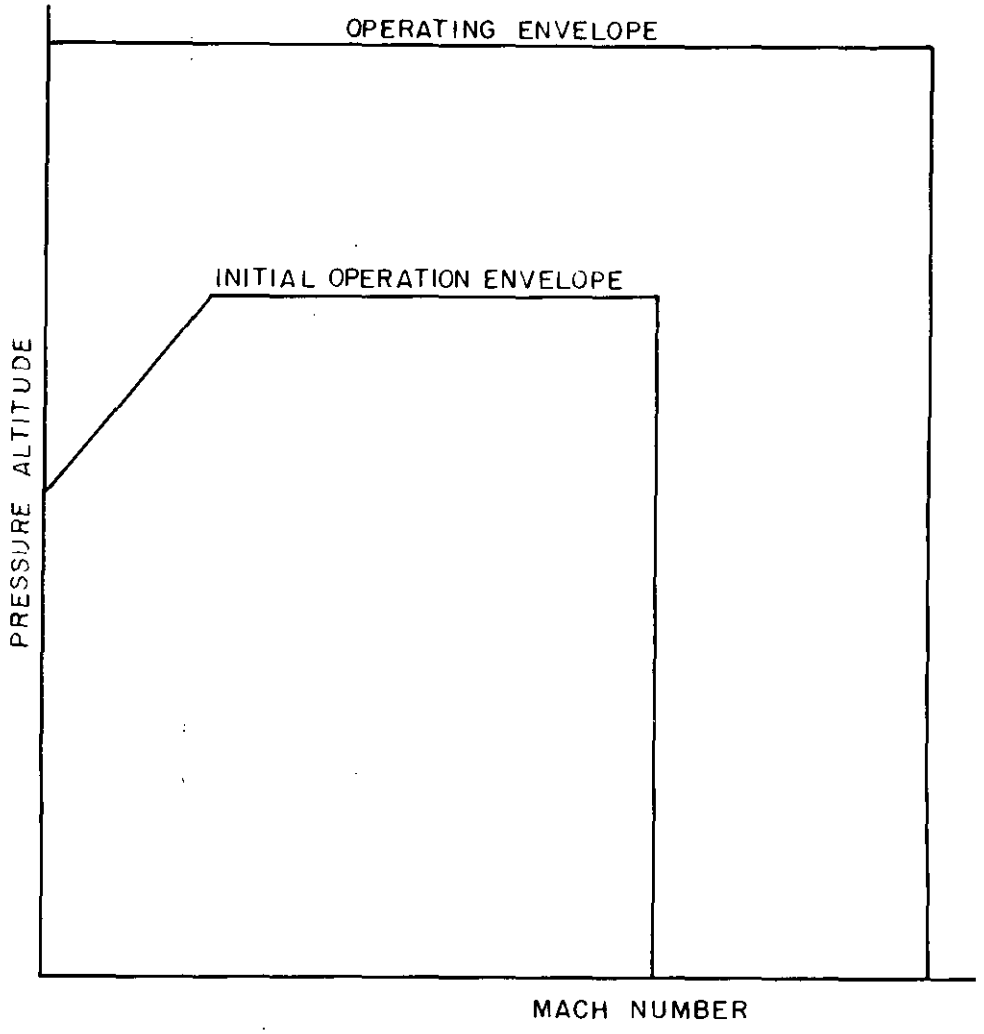


FIGURE 1. Starter Operating Envelope

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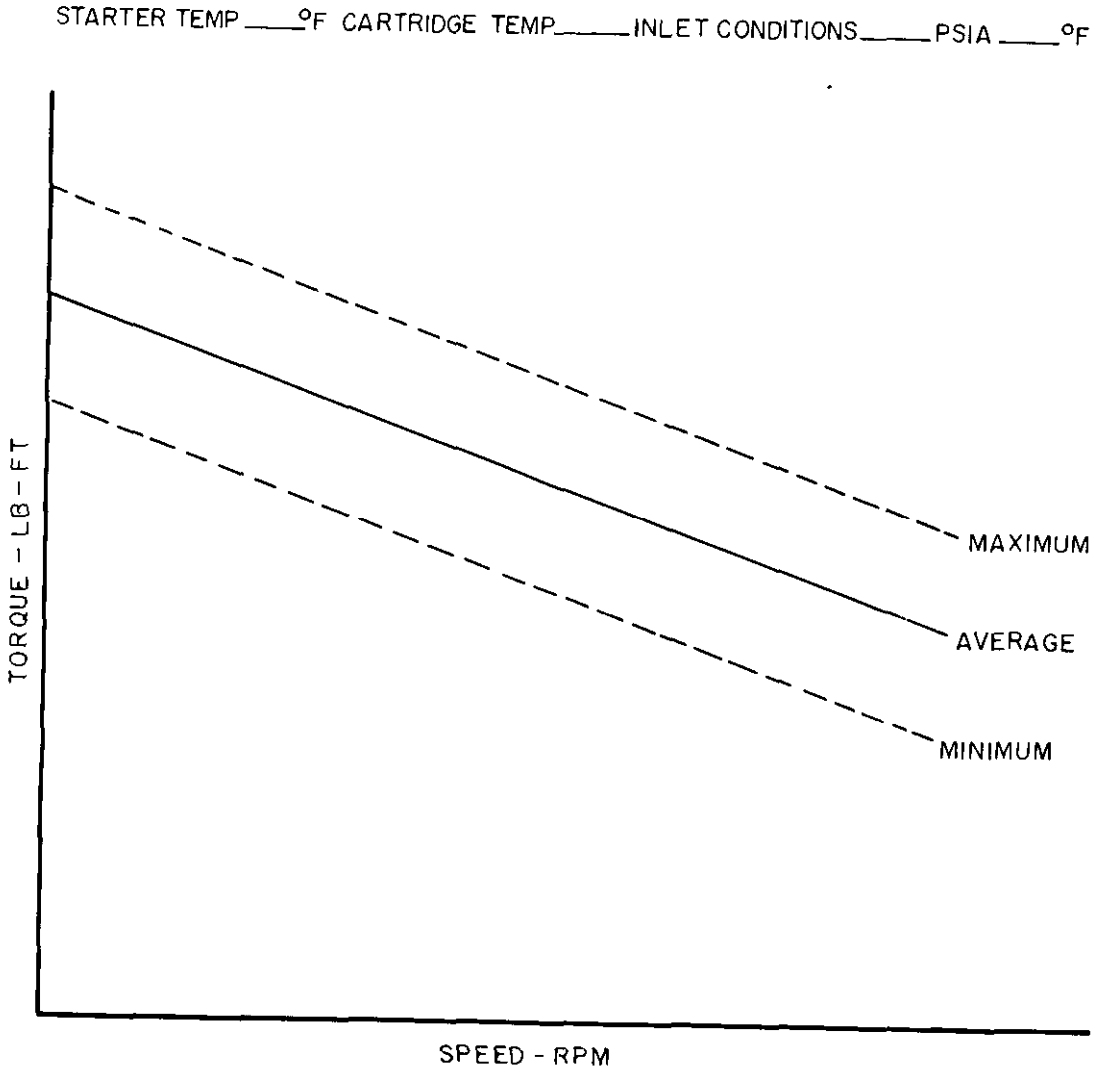


FIGURE 2. Variation of Sea Level Output Torque Versus RPM with Temperature and Inlet Conditions

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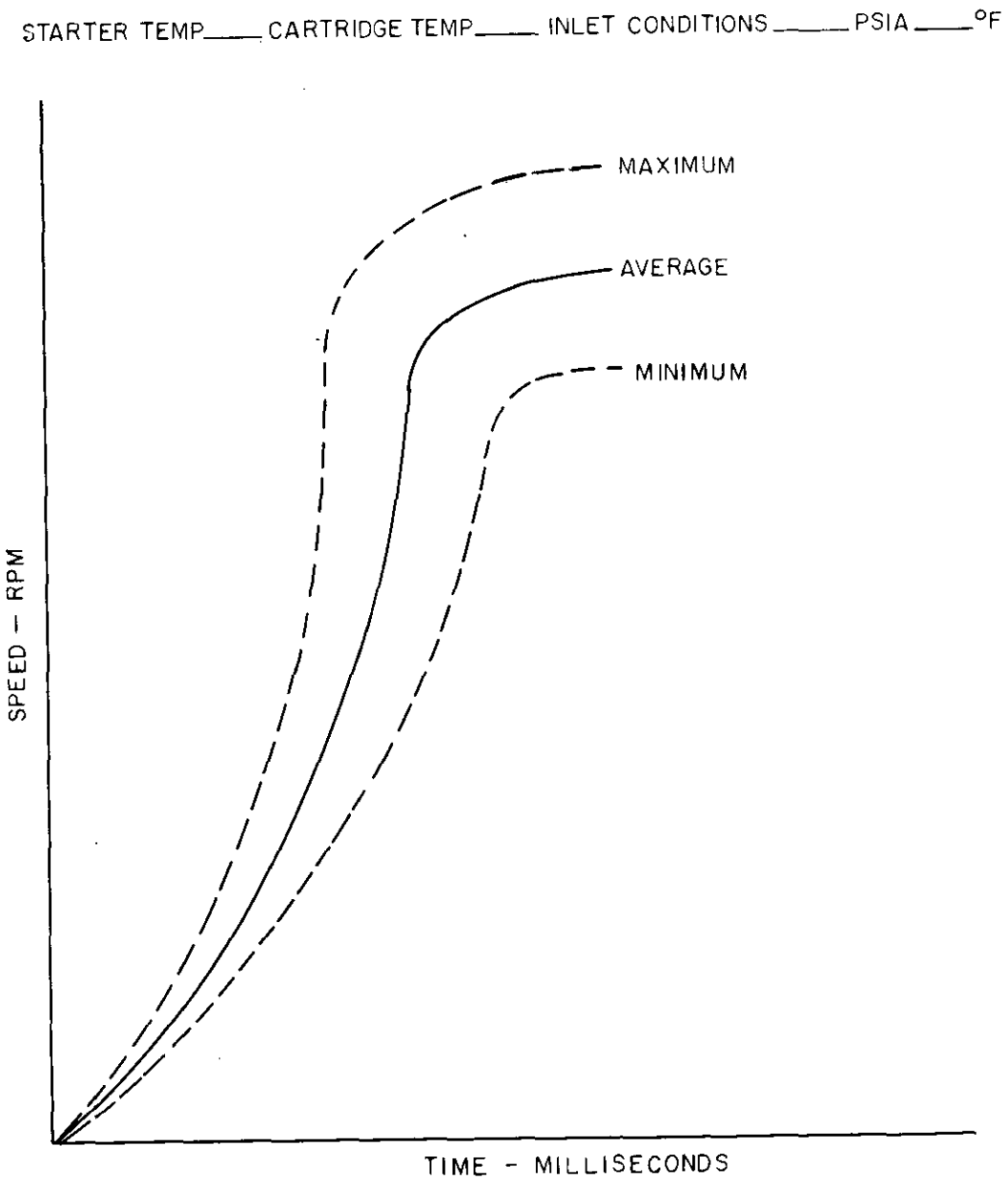


FIGURE 3. Variation of Starter RPM Versus Time (No-load Condition)

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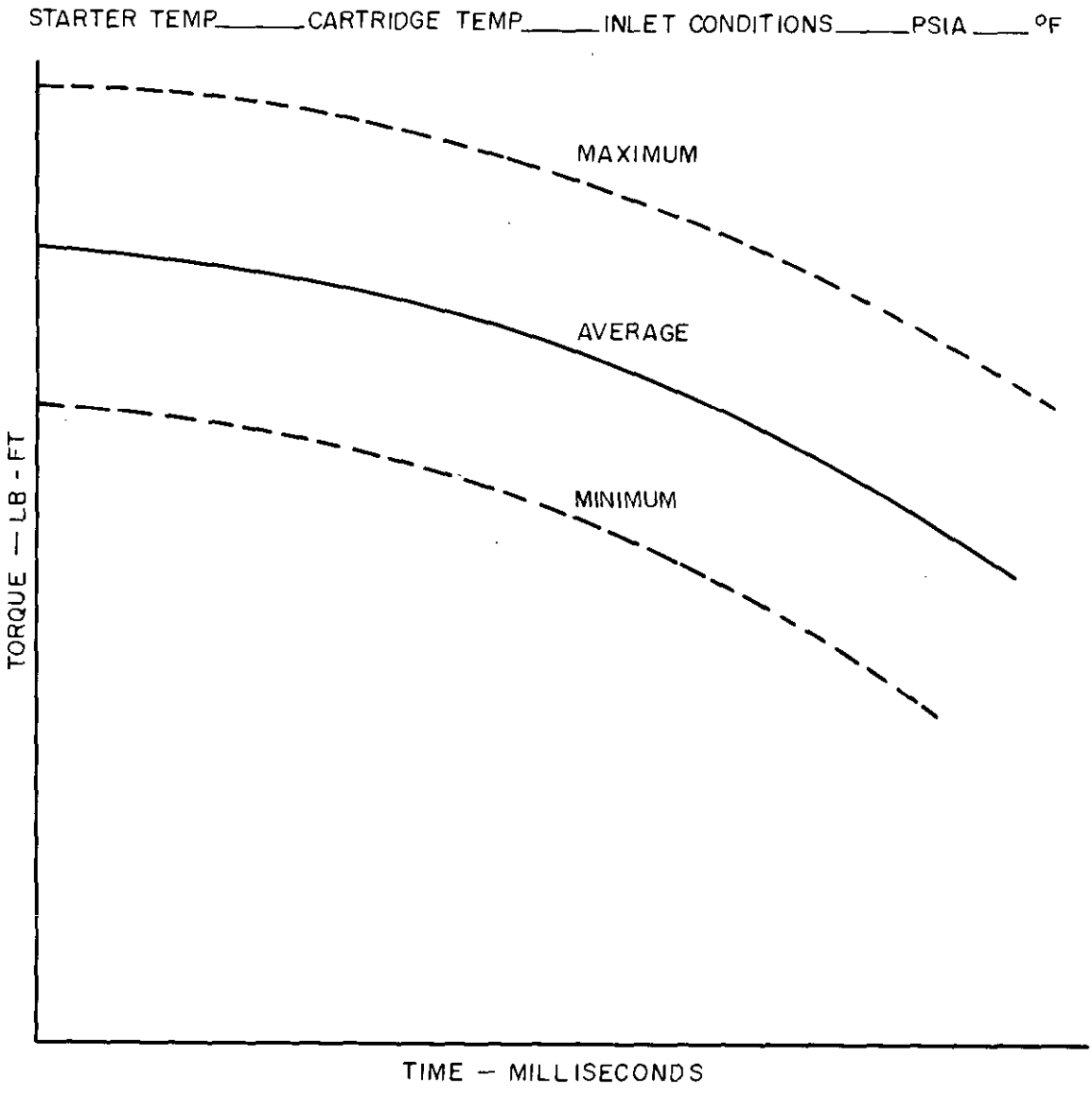


FIGURE 4. Variation of Starter Torque Versus Time with Temperature and Inlet Conditions at Sea Level

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STARTER TEMP_____CARTRIDGE TEMP_____INLET CONDITIONS_____PSIA ___°F

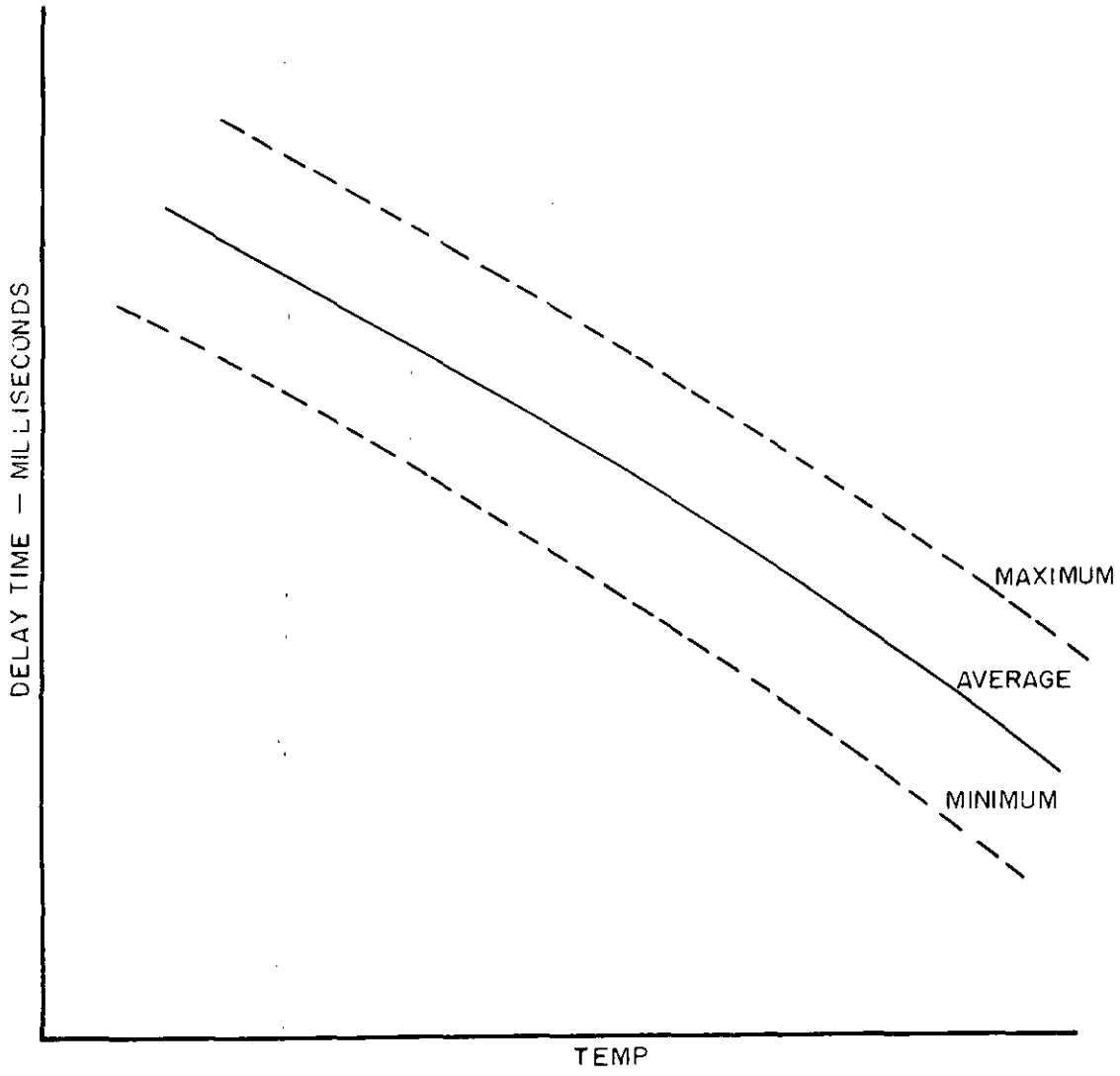


FIGURE 5. Variation of Delay Time with Temperature and Inlet Conditions at Sea Level

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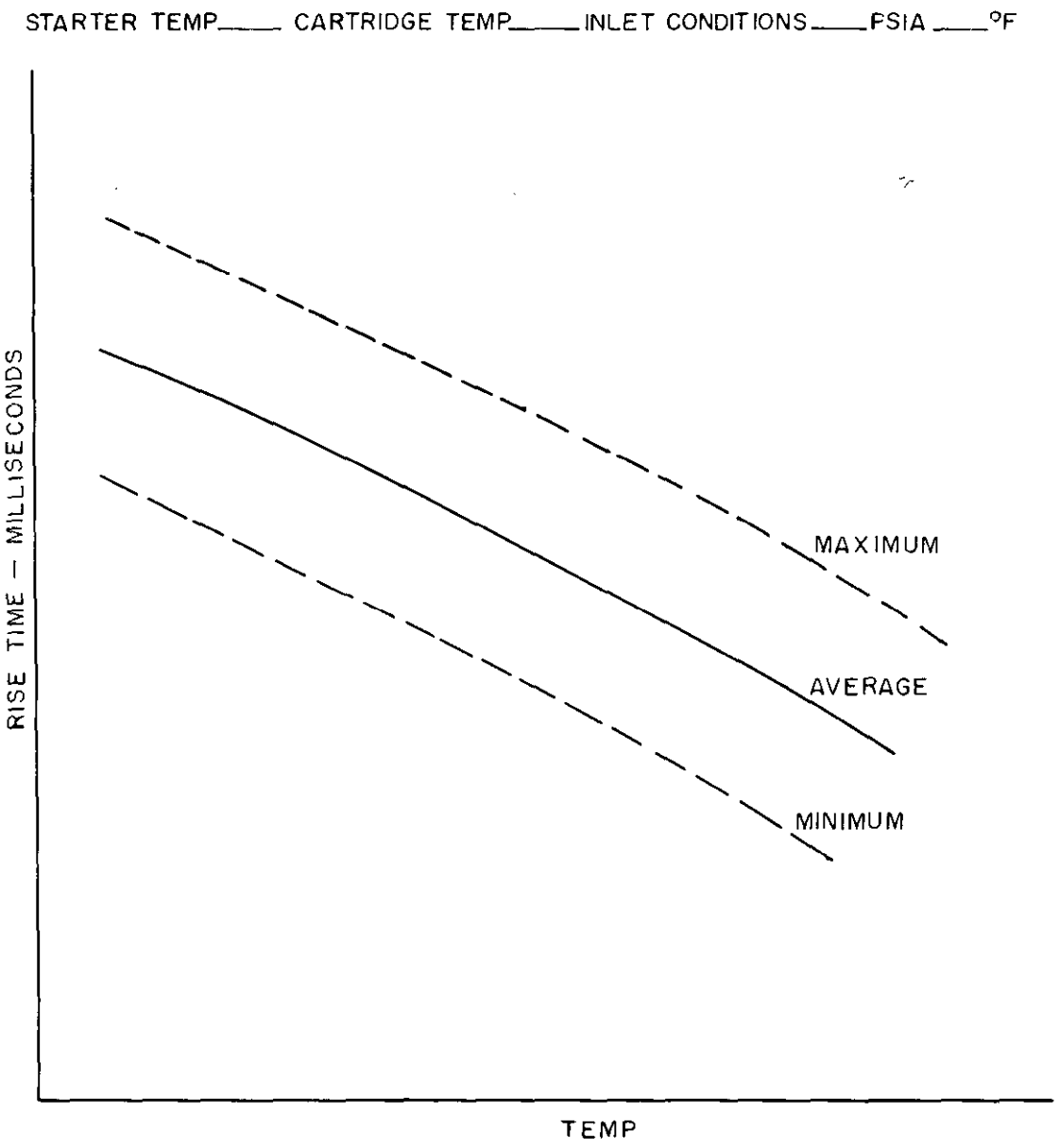


FIGURE 6. Variation of Rise Time with Temperature and Inlet Conditions at Sea Level

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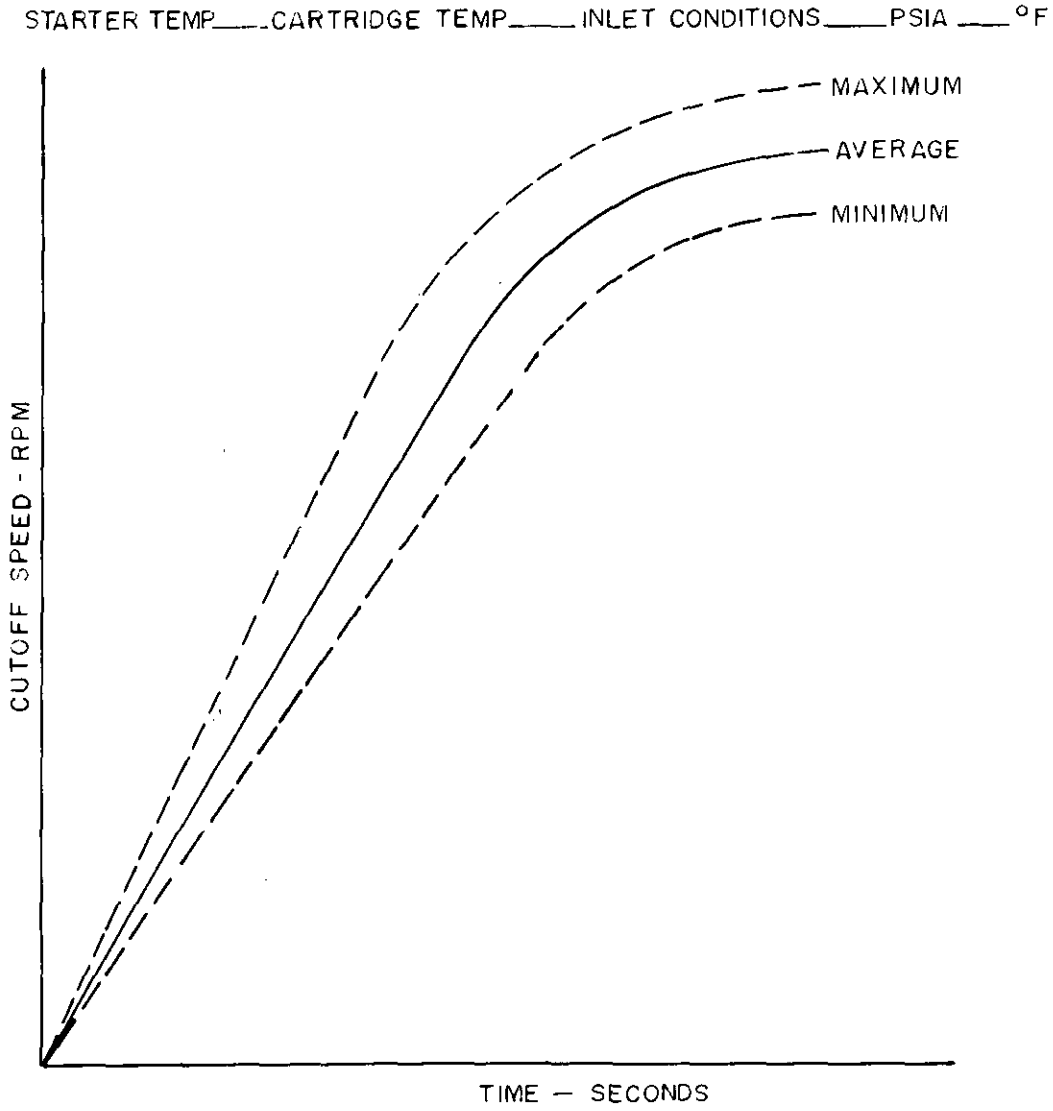


FIGURE 7. Variation of Starter Cutoff Speed Versus Time with *Temperature and Inlet Conditions at Sea Level*

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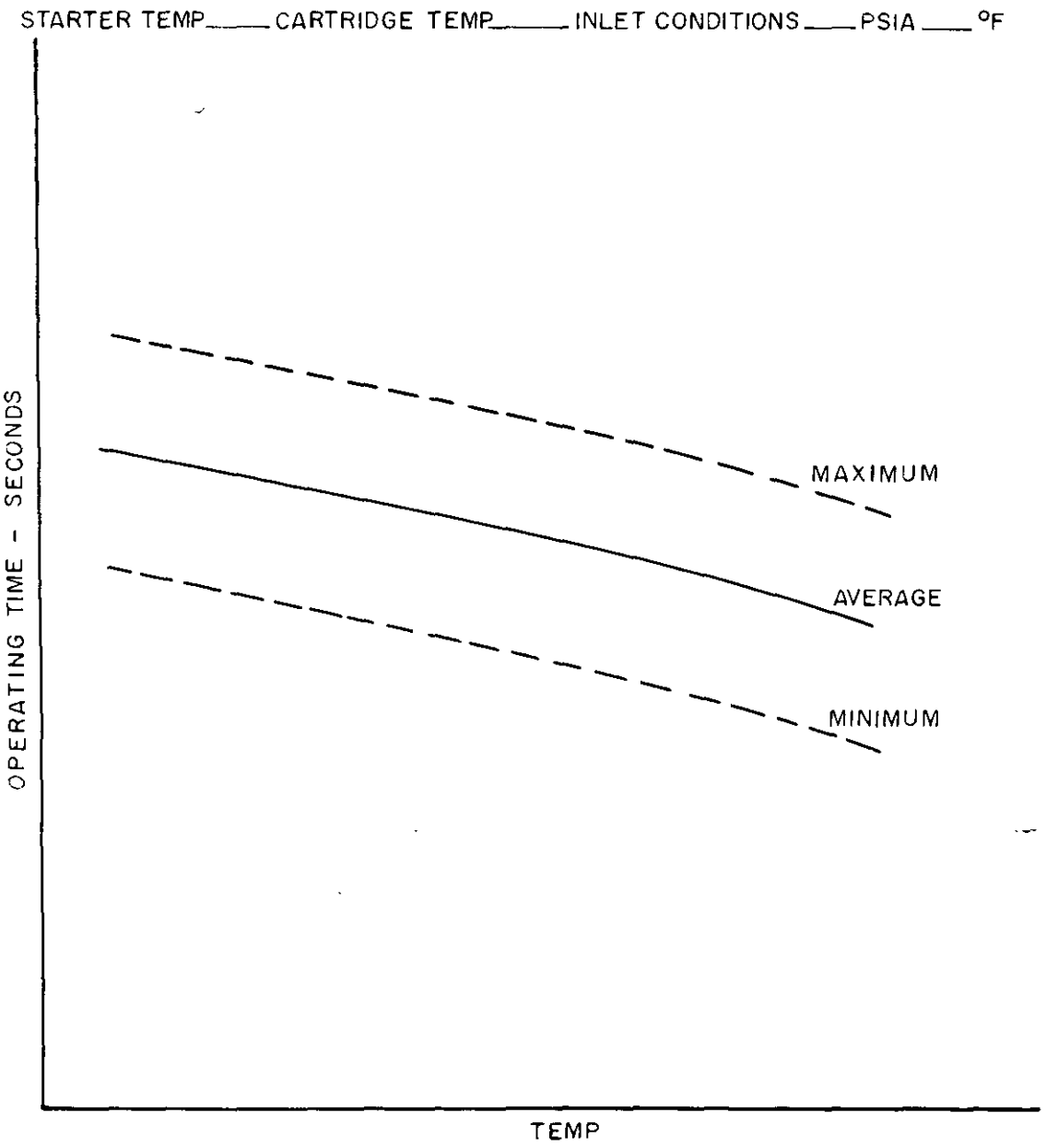


FIGURE 8. Variation of Operating Time with Temperature and Inlet Conditions at Sea Level

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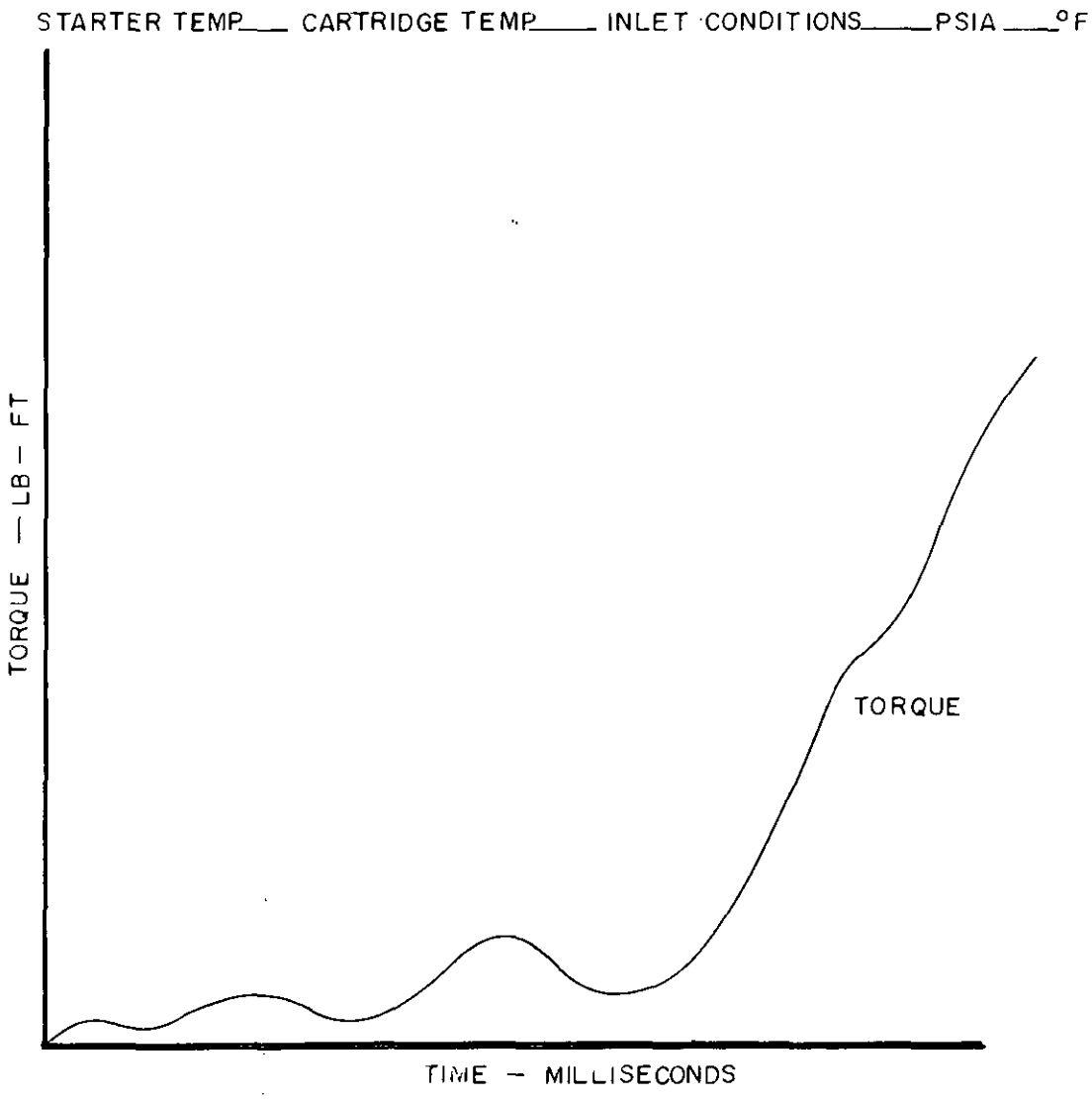


FIGURE 9. Typical Torque Transient During Actuation at Sea Level

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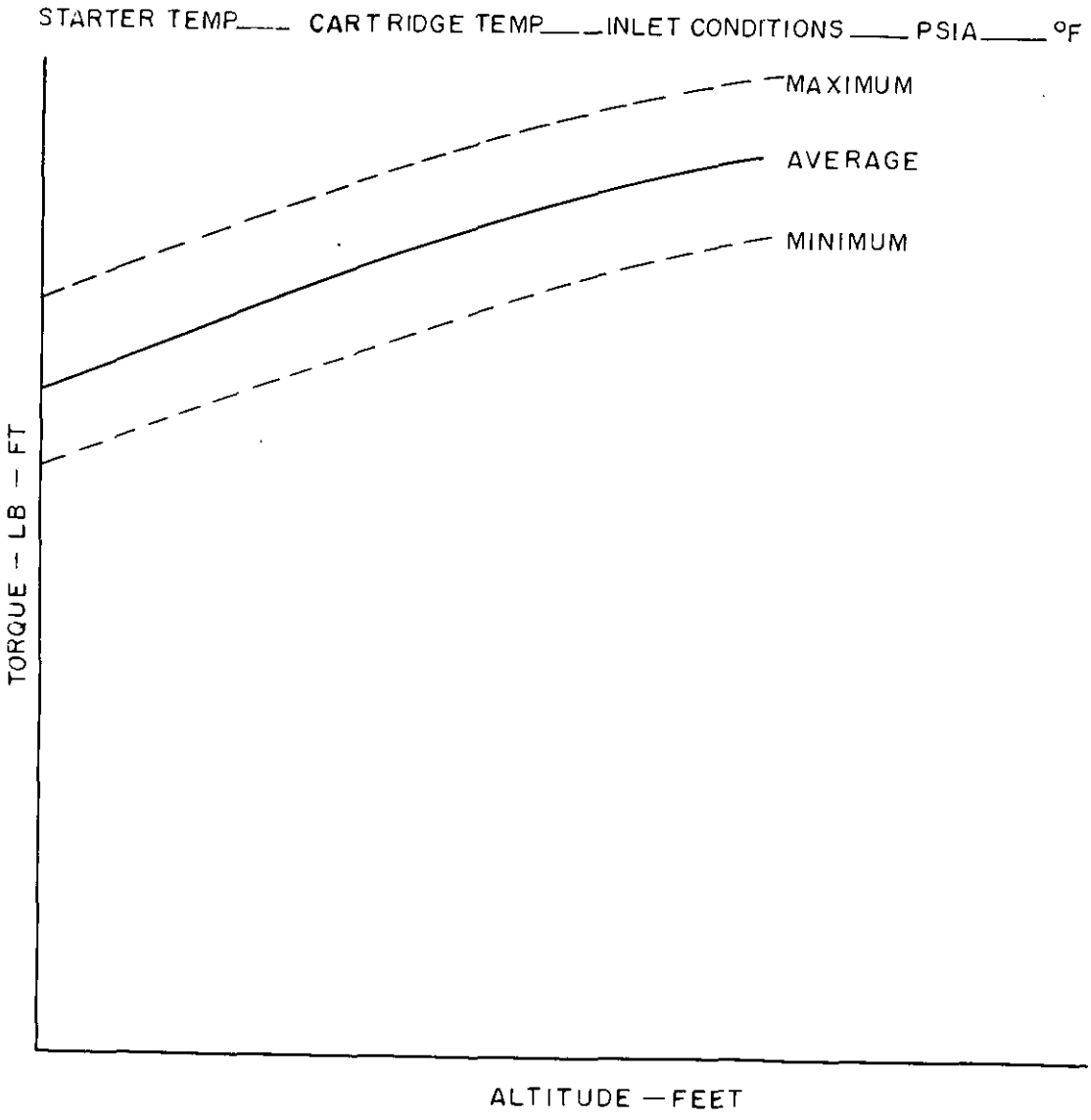


FIGURE 10. Variation with Altitude of Maximum Output Torque

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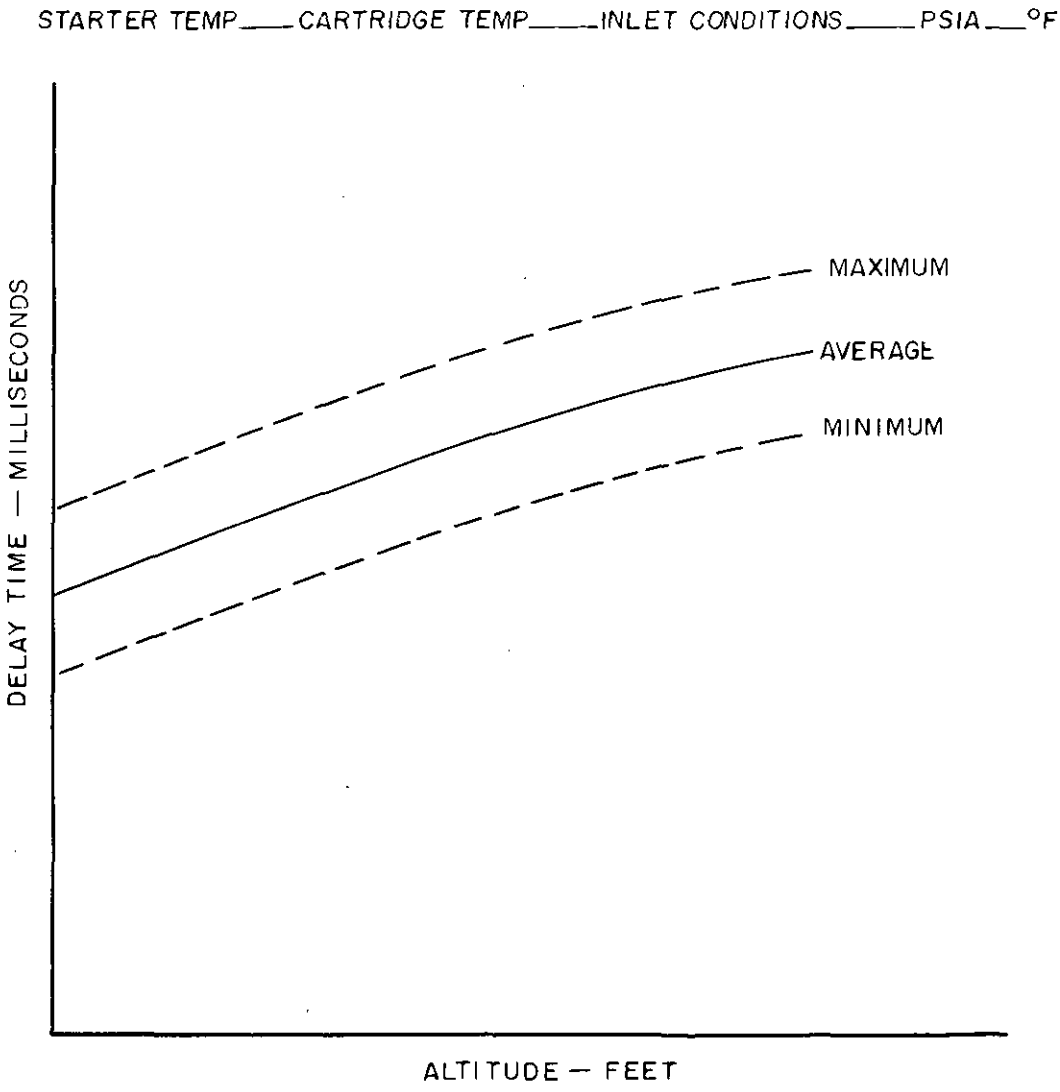


FIGURE 11. Variation with Altitude of Delay Time with Temperature at Sea Level

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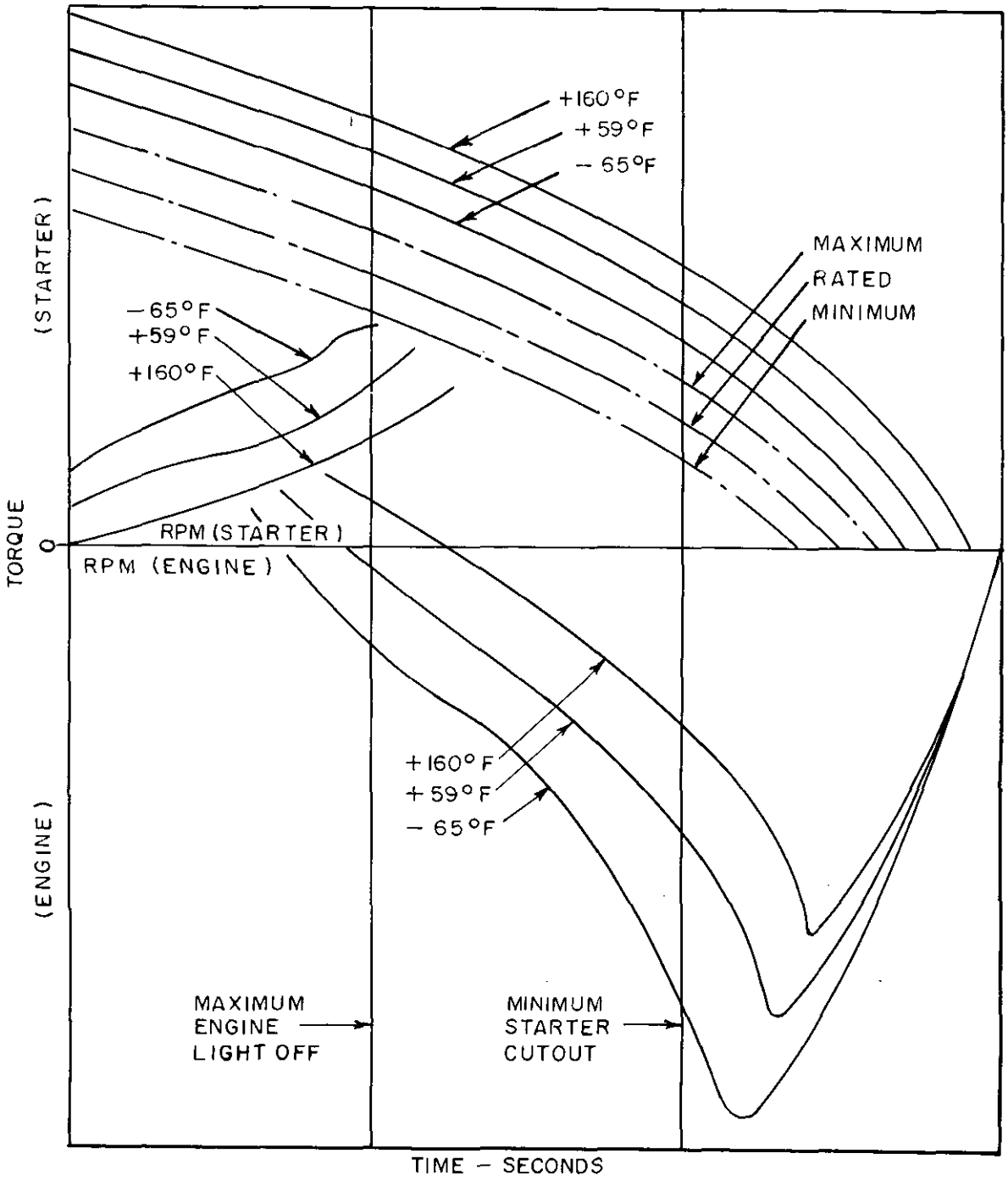


FIGURE 12. Starter-Engine Starting Characteristics

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3.8.3.3 Typical sea level starter rpm versus time with no load on the output shaft at the following conditions are shown on the following figures:

a. Cartridge operated

Starter and cartridge at -65°F	Figure_____
Starter and cartridge at +59°F	Figure_____
Starter and cartridge at +160°F	Figure_____

b. Pneumatic operated

Starter at -65°F, inlet conditions at maximum	Figure_____
Starter at +59°F, inlet conditions at rated	Figure_____
Starter at +160°F, inlet conditions at minimum	Figure_____

(These curves shall be drawn in accordance with figure 3.)

3.8.3.4 Curves showing the variation in starter performance with temperatures at sea level are shown on the following figures:

a. Cartridge operation with cartridge and starter conditioned at -65°F, +59°F, and +160°F:

Output torque	Figure_____
Delay time	Figure_____
Rise Time	Figure_____
Cutoff speed	Figure_____
Operating time	Figure_____

b. Pneumatic operation at maximum, rated, and minimum inlet conditions and starter conditioned to temperatures of -65°F, +59°F, and +160°F:

Output torque	Figure_____
Delay time	Figure_____
Rise time	Figure_____
Cutoff speed	Figure_____
Operating time	Figure_____

(These curves shall be drawn in accordance with figures 4 through 8.)

3.8.3.5 Curves showing typical starter output torque transients for cartridge operation with the starter and cartridge conditioned to -65°F, +59°F, and +160°F and for pneumatic operation using maximum, rated, and minimum inlet conditions are shown on figures_____ through _____.

(These curves shall be drawn in accordance with figure 9.)

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3.8.3.6 Curves showing variations in starter performance with altitude are shown on the following figures:

a. Maximum delay time and output torque for starter at -65°F, +59°F, and +160°F respectively with the air inlet conditions at:

-65°F Figure_____
+59°F Figure_____
+160°F Figure_____

(These curves shall be drawn in accordance with figures 10 and 11.)

3.8.4 Performance variation. The greatest values of maximum impact torque, maximum torque, and maximum breech pressure that occur in the temperature range of -65°F to +160°F are _____, _____, and _____, respectively. The inlet conditions at which the starter will be operated pneumatically are _____ psia and _____°F for maximum; _____ psia and _____ °F for rated conditons; and _____ psia and _____ °F for minimum conditions. Curves showing starter output torque at -65°F, +59°F, and 160°F (cartridge operated) and at maximum, rated, and minimum inlet conditions (pneumatic operated) superimposed on engine starting characteristics are shown on figures _____ through _____. (These curves shall be drawn in accordance with figure 12.)

3.9 Details of components. (In addition to the following, describe in detail all major components not specified below.)

3.9.1 Breech and gas transfer assembly

3.9.1.1.1 Breech pressures. The proof pressure of the breech is _____psia. The minimum yield pressure of the breech is _____psia.

3.9.1.1.2 Arming device. (Describe in detail the device utilized in arming the starter.)

3.9.1.2 Cartridge. (Reference the cartridge type and model or the cartridge manufacturer's model specification for the cartridge used with this starter.)

3.9.1.3 Nozzles. The following table presents data for the nozzles:

	Total Throat Area $in.^2$	Max Change in Area to Meet Table I & Table II Performance, $in.^2$	Flow Coefficient	Production Tolerance Area $in.^2$
Pneumatic Operation	_____	_____	_____	_____
Cartridge Operated	_____	_____	_____	_____

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3.9.1.4 Overpressure device

3.9.1.4.1 Actuation pressure. The actuation pressure of the overpressure device is _____ ± _____ psia.

3.9.1.4.2 Exit area. The exit area of the overpressure device is _____ $in.^2$.

3.9.1.4.3 Replacement and operation. (Describe in detail.)

3.9.1.5 Valves

3.9.1.5.1 Pneumatic control valve. The effective cross-sectional area of the pneumatic control valve is_____$in.^2$. The maximum flow rate is_____lb/sec at _____psia. Air pressure is regulated from_____to_____±_____psia. The actuation time from fully open to fully closed is_____±_____seconds (specify temperature effects), (specify the type of valve and applicable model specifications).

3.9.1.5.2 Gas pressure regulator. The maximum effective cross-sectional area of the gas pressure regulator valve is _____ $in.^2$. The maximum flow rate is _____ lb/sec at _____ psia. The minimum flow rate is _____ lb/sec at ____ psia. Gas pressure is regulated to _____ ± _____ psia. The minimum pressure required for actuation is _____ ± _____ psia.

3.9.1.6 Pneumatic air source. (Specify the pneumatic air source pressure, airflows, temperatures, and their limits.)

3.9.2 Energy conversion assembly

3.9.2.1 Turbine

3.9.2.1.1 Polar moment of inertia. The polar moment of inertia of the tubrine is _____ slug ft^2.

3.9.2.1.2.1 Maximum free-running speeds. The turbine maximum free-running speeds are as follows:

Cartridge Cycle		Pneumatic Cycle	
-65°F	_____rpm	-65°F	_____rpm
+59°F	_____rpm	+59°F	_____rpm
+160°F	_____rpm	+160°F	_____rpm

3.9.2.1.2.2 Minimum design yield speed. The turbine minimum yield speed is _____ rpm.

3.9.2.1.2.3 Turbine burst speed. The turbine minimum burst speed is _____rpm.

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3.9.2.1.2.4 Proof speed. The turbine proof speed is _____ rpm.

3.9.2.2 Other means of energy conversion. (Describe in detail any other means of energy conversion used.)

3.9.3 Speed reduction assembly

3.9.3.1 Gears. The gear ratio between the starter output shaft and the energy converter is _____ to 1. The gear backlash is _____ °.

3.9.4 Clutch and output assembly

3.9.4.1.1 Engaging clutch. The engage-disengage clutch shall be set to engage the engine from 0 to _____ rpm ± _____ rpm and to disengage the starter from _____ rpm to _____ rpm ± _____ rpm.

3.9.4.1.2 Slip clutch. The maximum breakaway torque for the slip clutch is _____ ± _____ lb-in. (Specify for all cartridge temperatures and turbine inlet conditions.)

3.9.4.2 Output shaft. Based upon infinite fatigue life, the maximum torque the starter output shaft can withstand is _____ ± _____ lb-in. The direction of rotation when facing the starter output shaft end is _____ clockwise.

3.9.4.2.1 Shear section. The applied torque value at which the shaft will shear is _____ ± _____ lb-in. The number of cycles to failure is _____. The rate of loading used was _____ lb-in./sec.

3.9.4.3 Mounting flange and drive. The engine accessory drive type number used is _____.

3.9.5 Control system

3.9.6 Duct system

3.9.6.1 Air intake duct. The effective cross-sectional area of the air intake duct is _____ $in.^2$. The length is _____ inches. It will handle an air intake rate of _____ lb/sec at a pressure of _____ psia.

3.9.6.2 Exhaust duct. The effective cross-sectional area of the exhaust system duct is _____ $in.^2$. The length is _____ inches. The duct will handle a total gas flow of _____ lb/sec.

3.9.7 Power system

3.9.7.1.4 Starter cartridge circuit. The maximum resistance through the starter cartridge circuit is _____ ohms.

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3.9.7.2 Other power systems. (Specify any other type of power required by the starter.)

3.9.8 Lubrication system

3.9.8.1 Oil supply. The average oil consumption rate of the starter is _____ oz/cycle. The minimum amount of oil required in the reservoir for satisfactory starter operation is _____ oz. The amount of oil required to fill the reservoir is _____ oz.

3.9.8.2 Oil loss. The starter maximum oil loss rate is _____ oz/cycle.

3.9.8.3 Oil flow and heat rejection. The oil flow rate is _____ lb/sec. The heat rejection rate is _____ BTU/hr.

3.9.9 Other components. (Describe any starter components not otherwise specified. List their proof and burst pressures if components are normally subjected to gas pressure.)

3.11 Weight of starter

3.11.1 With cartridge. The weight of the starter with cartridge is _____lb.

3.11.2 Without cartridge. The weight of the starter without cartridge is _____ lb.

3.13 Drawings and data. The following_____(insert manufacturer's corporate names) drawings and data form a part of this specification:

a. Starter installation drawing (list drawing number)

b. Electrical installation connection diagrams (list drawing numbers)

c. Starter components and assembly drawing (list drawing number).

3.14 Identification of product. The starter shall be marked for identification in accordance with drawing number _____ (list drawing numbers).

3.17 General additional requirements. (The starter manufacturer may specify as subparagraphs under this number and heading any additional information or requirements which are not covered by the required statements herein or the deviations taken to the basic specification.)

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4. QUALITY ASSURANCE PROVISIONS

4.1 The requirements for quality assurance shall be as specified in MIL-S-27266.

4.3.6 Test apparatus

4.3.6.3 Test flywheels

4.3.6.3.1 Flywheel A. The polar moment of inertia of flywheel A is _____ slug ft^2.

4.3.6.3.2 Flywheel B. The polar moment of inertia of flywheel B is _____ slug ft^2.

4.4.1 Preproduction test method. Preproduction tests as specified in MIL-S-27266 shall be conducted on sample starters in accordance with the preproduction test method as described in document _____ (list document number and type or model of starter) and approved by the procuring activity.

4.5 Quality conformance tests. Quality conformance tests as specified in MIL-S-27266 and conducted in accordance with the following quality conformance test method as approved by the procuring activity shall be accomplished on production starters and starters submitted for preproduction tests.

4.5.4 Quality conformance test method. (Describe the complete quality conformance test method herein.)

5. PREPARATION FOR DELIVERY. The requirements for preparation of delivery shall be as specified in MIL-S-27266.

6 NOTES

6.3 Definitions and symbols. The definitions and symbols used in this specification shall be as specified in MIL-S-27266.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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